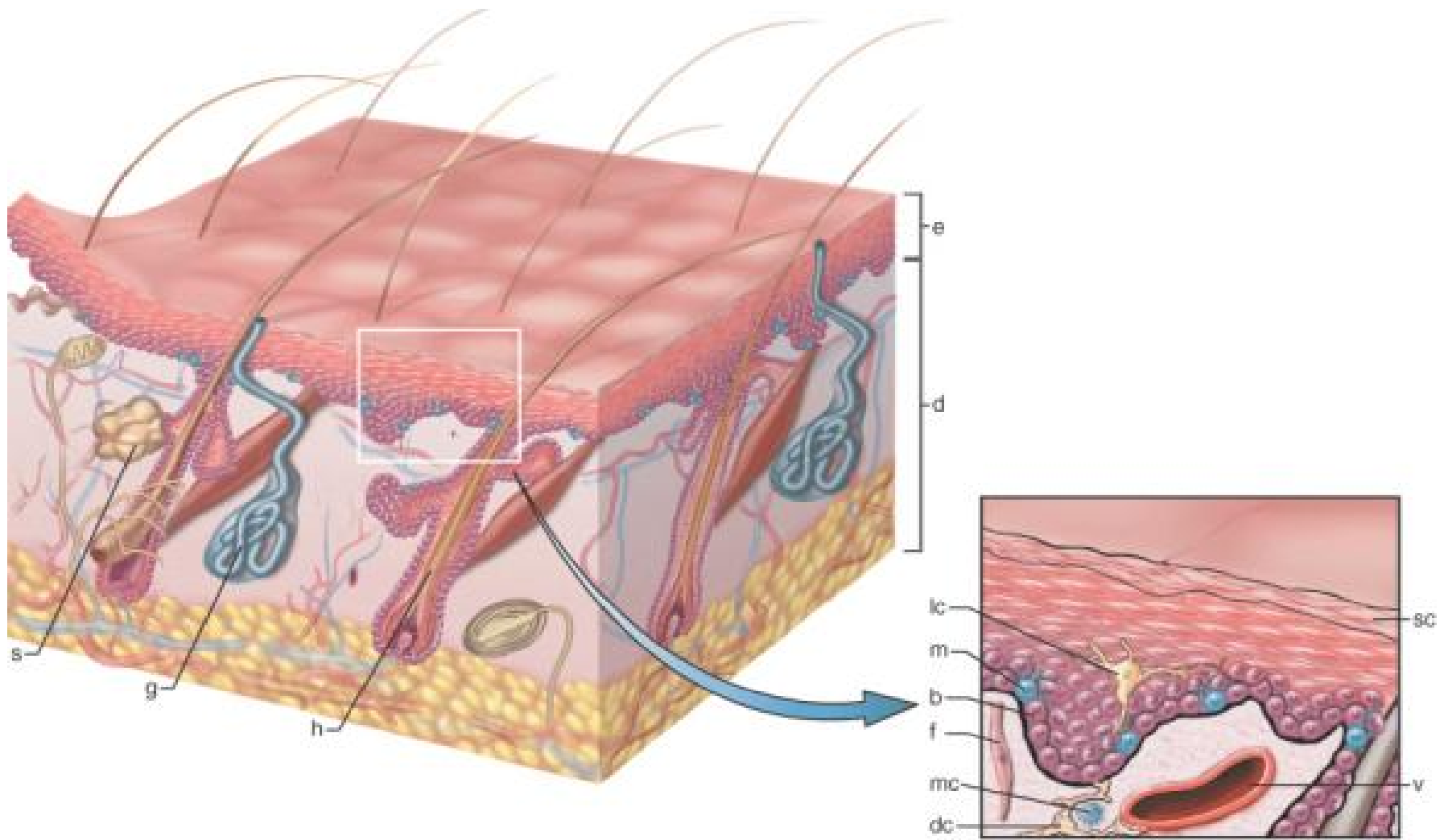


Diseases of the Skin



MELESE A.(MD,PATHOLOGIST)



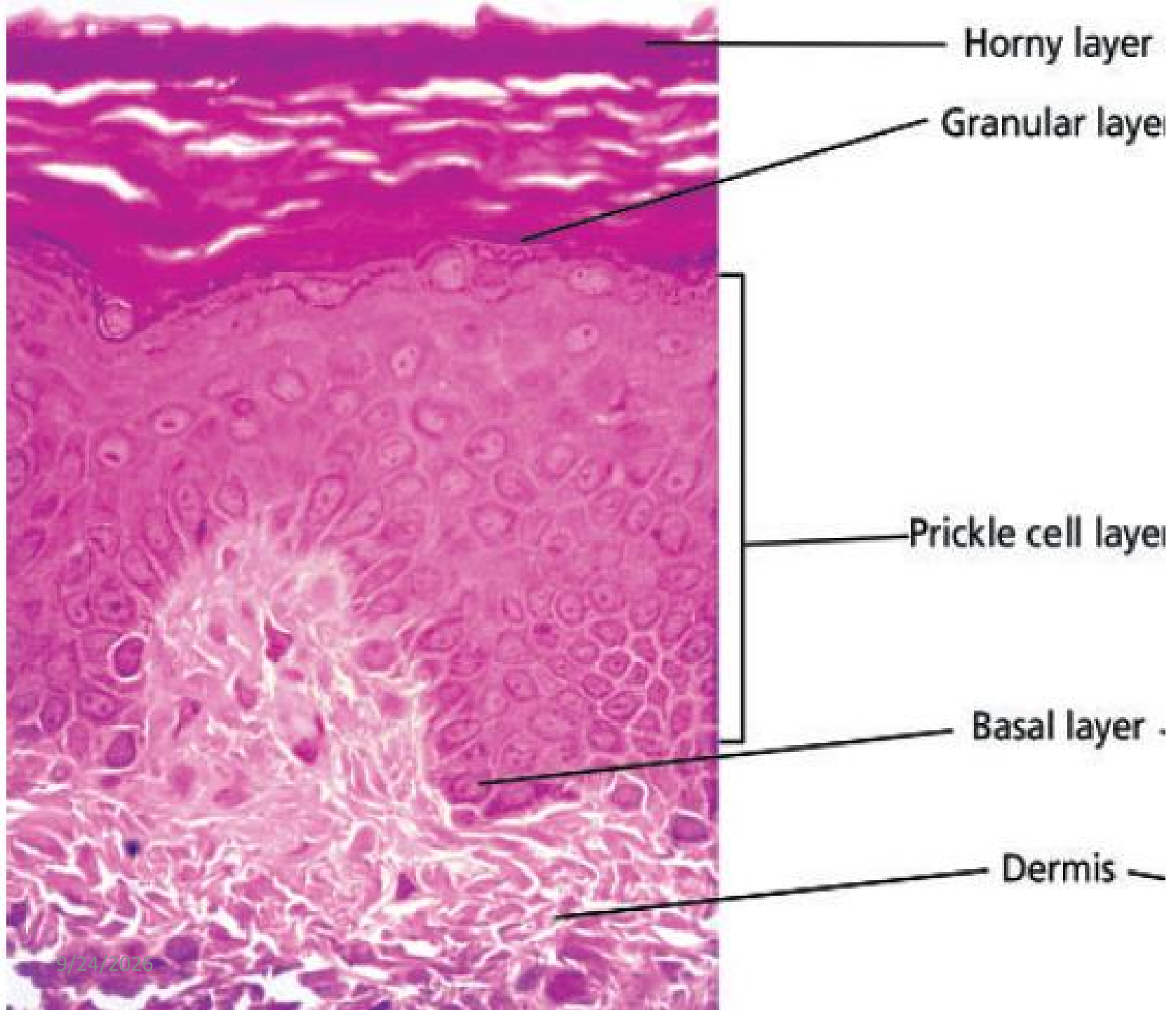
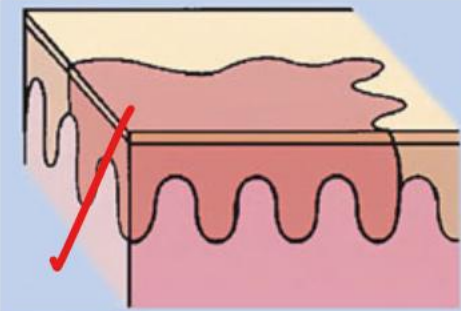
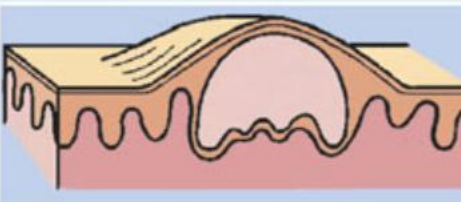
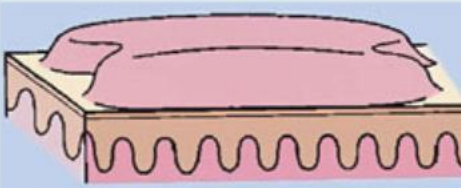
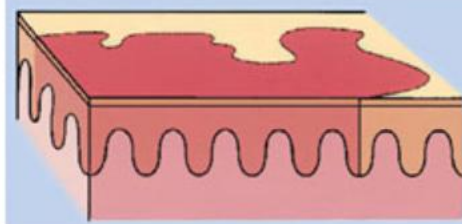
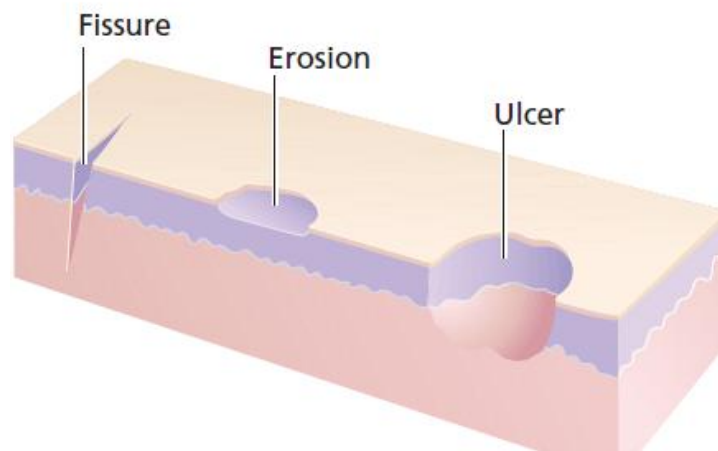
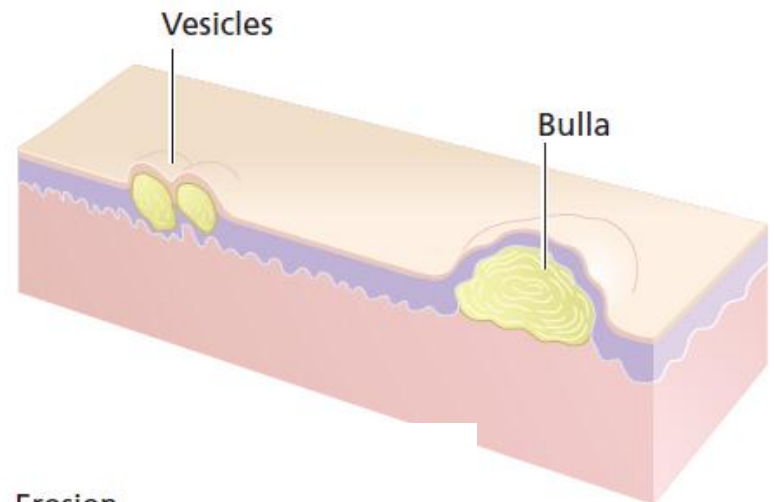
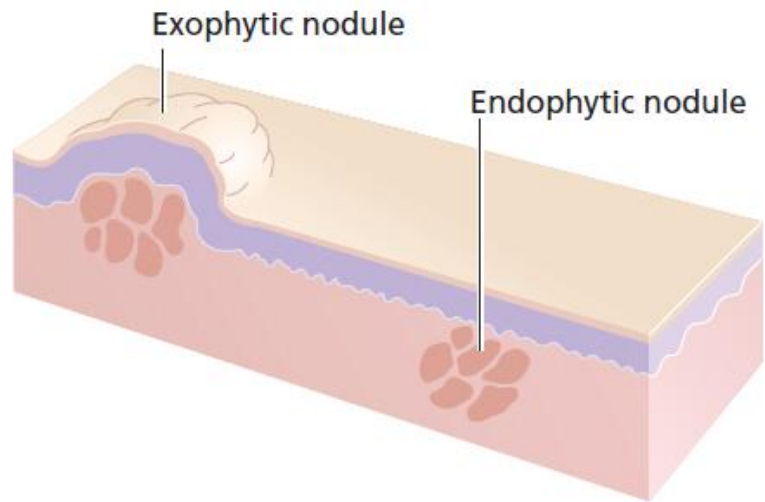
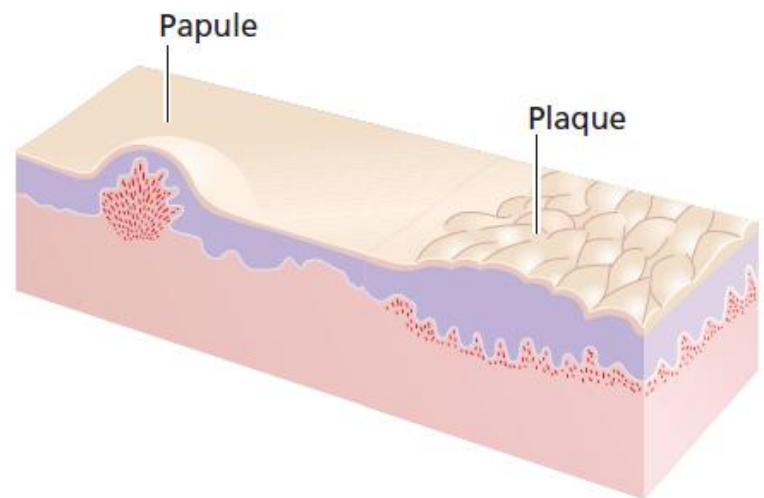
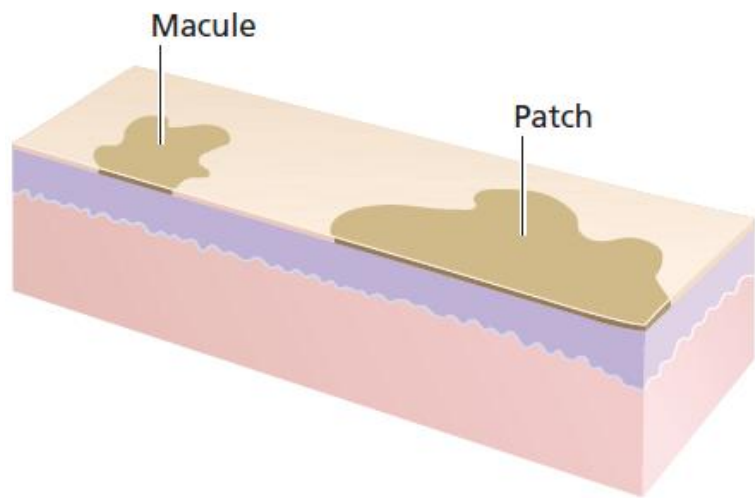


Table 25.1 Nomenclature of Skin Lesions

Definition	
Macroscopic Lesions	
Excoriation	Traumatic lesion breaking the epidermis and causing a raw linear defect (i.e., deep scratch); often self-induced.
Lichenification	Thickened, rough skin (similar to lichen on a rock); usually the result of repeated rubbing.
Macule, Patch	Circumscribed, flat lesion distinguished from surrounding skin by color. Macules are 5 mm in diameter or less; patches are greater than 5 mm.
Onycholysis	Separation of nail plate from nail bed.
Papule, Nodule	Elevated dome-shaped or flat-topped lesion. Papules are 5 mm or less across; nodules are greater than 5 mm in size.
Plaque	Elevated flat-topped lesion, usually greater than 5 mm across (may be caused by coalescent papules).
Pustule	Discrete, pus-filled, raised lesion.
Scale	Dry, horny, plate-like excrescence; usually the result of imperfect cornification.
Vesicle, Bulla, Blister	Fluid-filled raised lesion 5 mm or less across (vesicle) or greater than 5 mm across (bulla). Blister is the common term for either lesion.
Wheal	Itchy, transient, elevated lesion with variable blanching and erythema formed as the result of dermal edema.

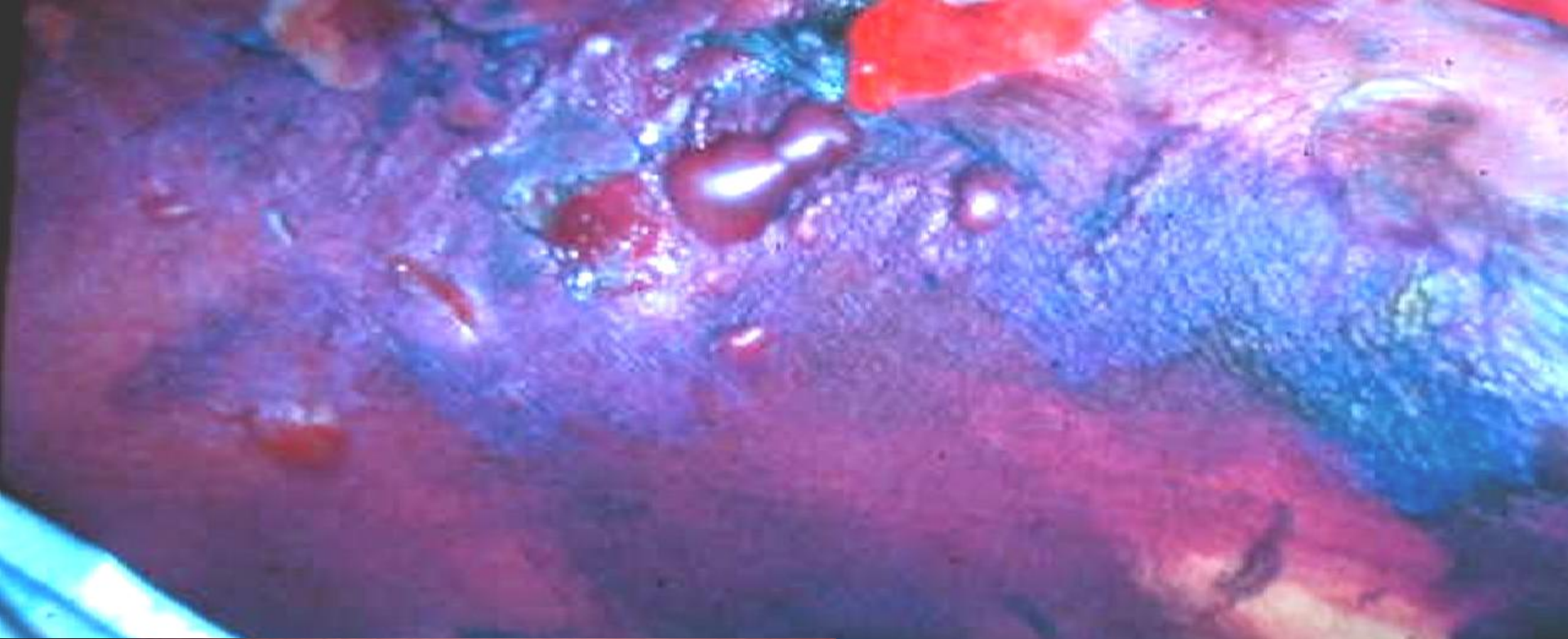
PRIMARY LESION	DEFINITION	MORPHOLOGY	EXAMPLES
Macule	Flat, circumscribed skin discoloration that lacks surface elevation or depression		Café-au-lait Vitiligo Freckle Junctional nevi Ink tattoo
Papule	Elevated, solid lesion <0.5 cm in diameter		Acrochordon (skin tag) Basal cell carcinoma Molluscum contagiosum
Plaque	Elevated, solid "confluence of papules" (>0.5 cm in diameter) that lacks a deep component		Bowen's disease Mycosis fungoides Psoriasis Eczema Tinea corporis

Patch	Flat, circumscribed skin discoloration; a very large macule		Port wine stain Vitiligo
Nodule	Elevated, solid lesion >0.5 cm in diameter; a larger, deeper papule		Rheumatoid nodule Tendon xanthoma Erythema nodosum Lipoma Metastatic carcinoma



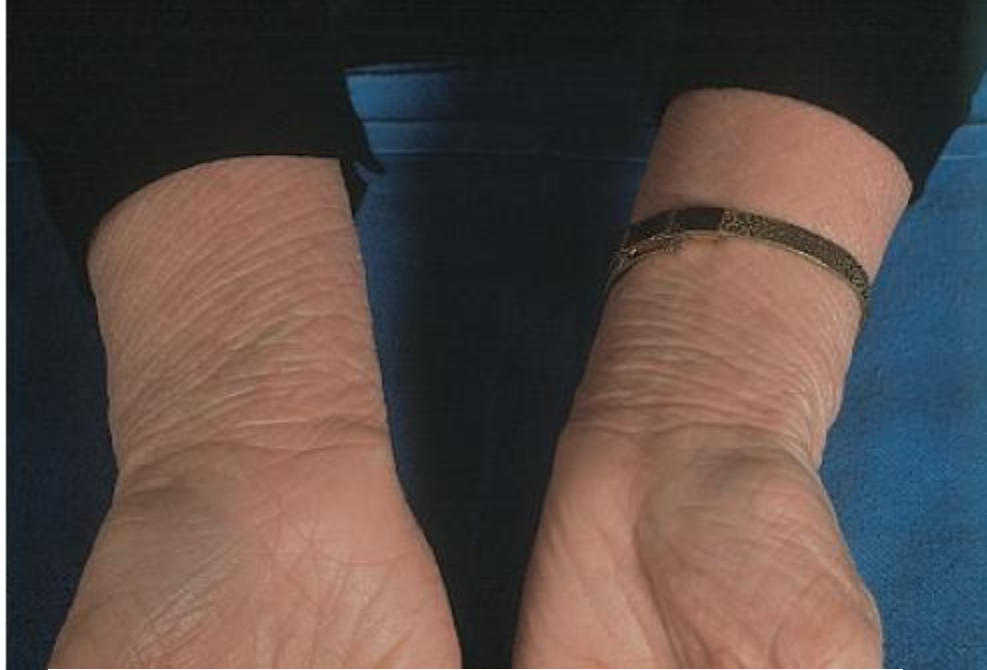


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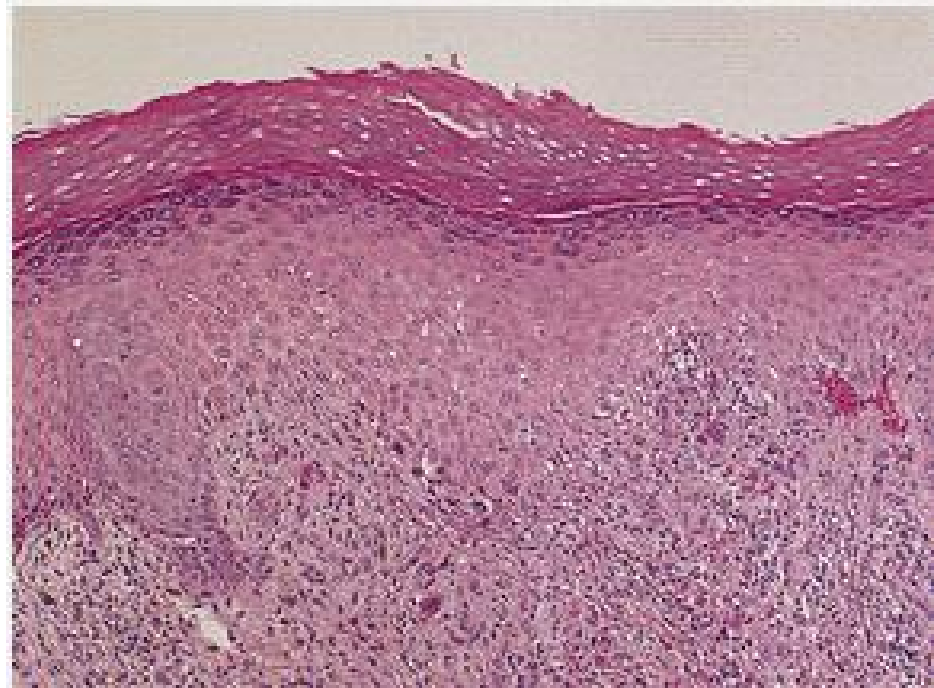
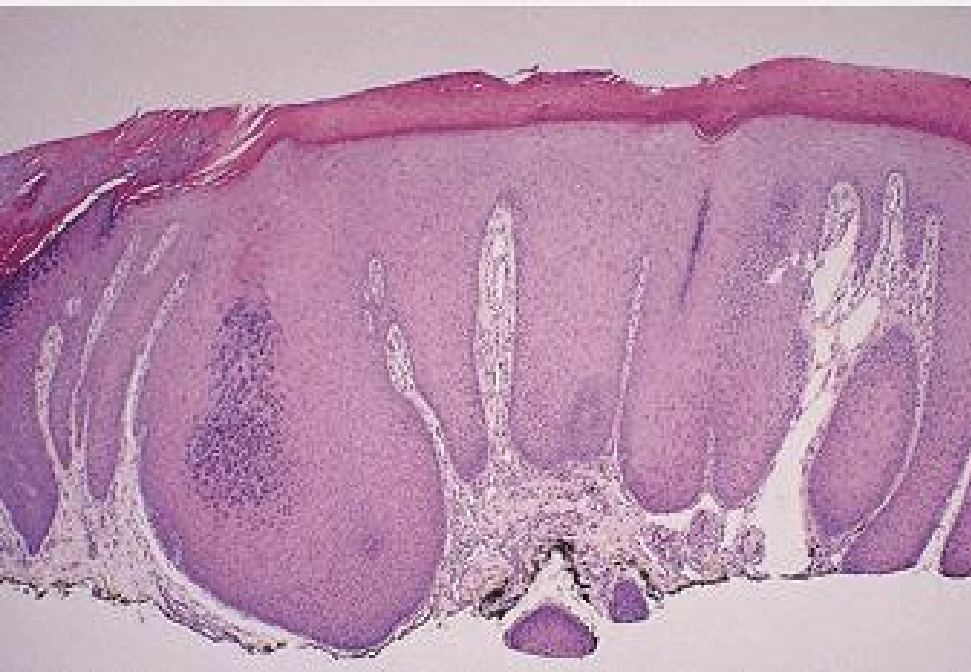
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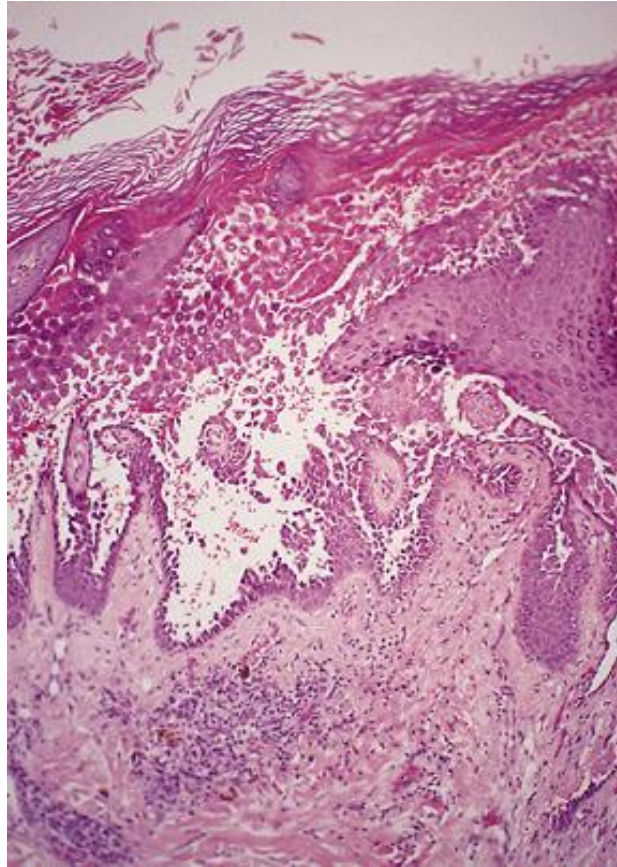


DEFINITIONS OF MICROSCOPIC TERMS

- **Hyperkeratosis**
 - Thickening of the **stratum corneum**, often associated with a qualitative abnormality of the keratin.
- **Parakeratosis**
 - Modes of keratinization characterized by the retention of the nuclei in the stratum corneum. On mucous membranes, parakeratosis is normal.
- **Hypergranulosis**
 - Hyperplasia of the stratum granulosum, often due to intense rubbing.
- **Acanthosis**
 - Diffuse epidermal hyperplasia.

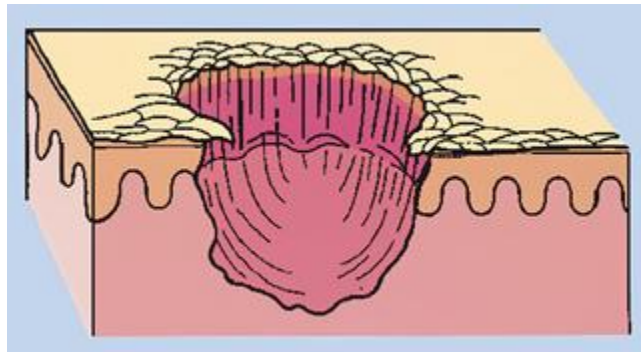
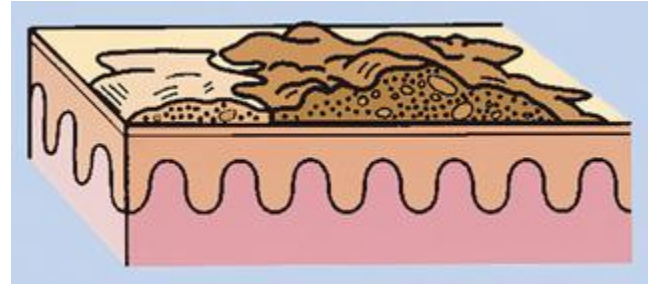
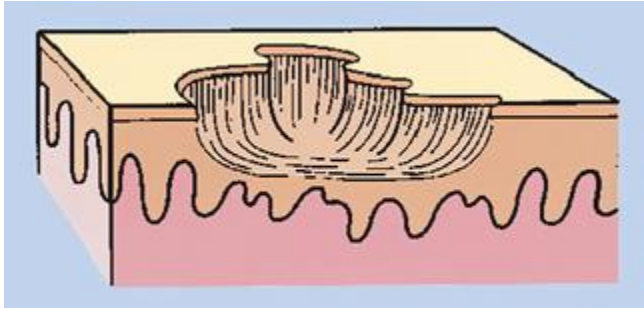


- **Papillomatosis**
 - Surface elevation caused by hyperplasia and enlargement of contiguous dermal papillae.
- **Dyskeratosis**
 - Abnormal keratinization occurring prematurely within individual cells or groups of cells below the stratum granulosum.
- **Acantholysis**
 - Loss of intercellular connections resulting in loss of cohesion between keratinocytes.
- **Spongiosis**
 - Intercellular edema of the epidermis.
- **Hydropic swelling (ballooning)**
 - Intracellular edema of keratinocytes, often seen in viral infections.



Microscopic Terms Cont'd

- **Exocytosis**
 - Infiltration of the epidermis by inflammatory or circulating blood cells.
- **Erosion**
 - Discontinuity of the skin exhibiting **incomplete** loss of the epidermis.
- **Ulceration**
 - Discontinuity of the skin exhibiting **complete** loss of the epidermis and often of portions of the dermis and even subcutaneous fat.
- **Vacuolization**
 - Formation of vacuoles within or adjacent to cells; often refers to **basal cell-basement membrane** zone area.
- **Lentiginous**
 - Referring to a linear pattern of melanocyte proliferation within the epidermal basal cell layer.
 - Lentiginous melanocytic hyperplasia can occur as a reactive change or as part of a neoplasm of melanocytes.



Inflammatory Disorders of the Skin

- ❖ This is a large category where lesions are classified by the
 - ❑ **predominant inflammatory cell type** and
 - ❑ the **pattern of involvement**.

Acute Inflammatory Dermatoses

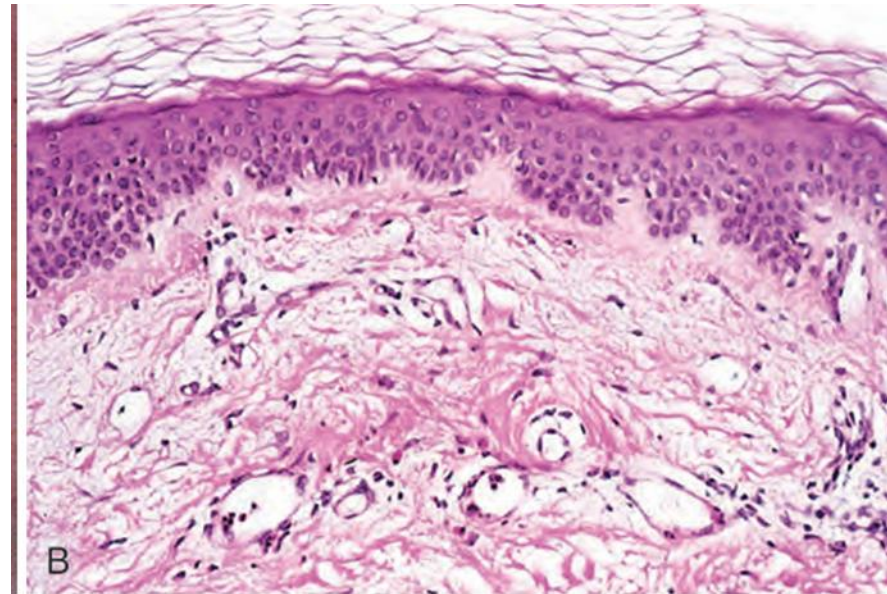
- ❖ Often of sudden onset
- ❖ characterized by
 - fluid accumulation (edema)
 - vasodilation
 - transient inflammatory infiltrate.
 - often rich in **neutrophils** or **eosinophils**.
- ❖ They include
 - Urticaria
 - Acute eczematous dermatitis (spongiotic dermatitis)
 - Erythema multiforme

Urticaria

- ❖ The result of **antigen-induced** release of vasoactive mediators from mast cells.
- ❖ Usually due to **type I** hypersensitivity reaction
 - In response to various allergens (pollen, food, drugs, insect venom)
- ❖ Pruritic, erythematous, raised wheals that appear and disappear rapidly.

Morphology:

- **Dermal Edema:** The hallmark. Separation of collagen fibers due to fluid accumulation.
- **Sparse Perivascular Infiltrate:** Composed of lymphocytes and **eosinophils**.

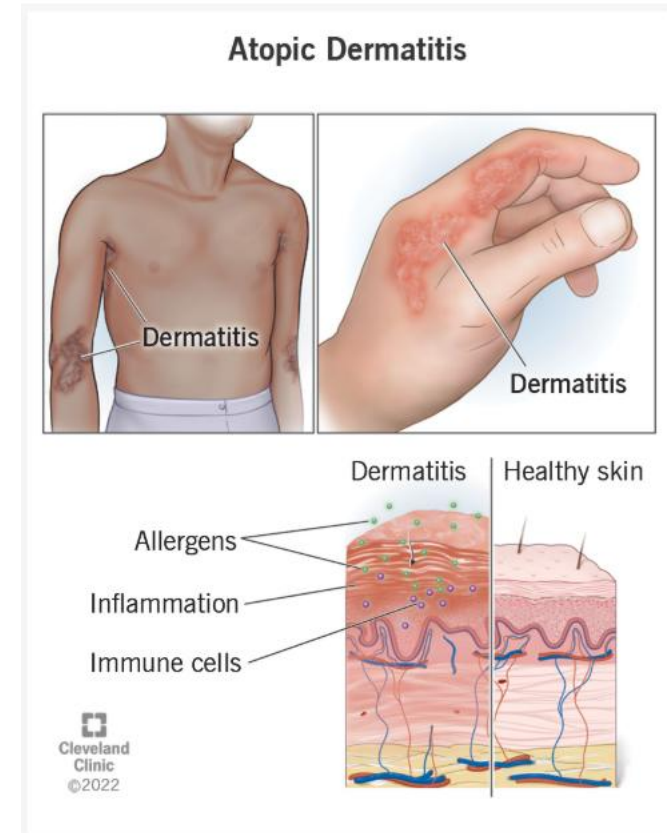


Acute Eczematous Dermatitis

- ❖ A group of diseases which are characterized by **vesicle formation** at early stages and **thickening and lichenification** at late stages

Etiology:

- ❖ **Allergic Contact Dermatitis:** Type IV (delayed) hypersensitivity to an external allergen (e.g., poison ivy, nickel).
- ❖ **Atopic Dermatitis:** Complex genetic and immune dysregulation, often associated with elevated IgE and a personal/family history of asthma/hay fever.
- ❖ **Irritant Contact Dermatitis:** Direct chemical damage to the skin barrier



Clinical features

- ❖ Itchy, erythematous, papulovesicular rash that can weep, scale, and crust.
- ❖ It has three phases - acute, subacute and chronic
 - **Acute** phase shows **vesicles** which rupture and ooze exudative material.
 - **Subacute** phase - shows crusting of the ruptured vesicles. This may subside.
 - **Chronic** phase - shows thickening of the skin with prominent skin markings (lichenification)

Histology

Varies depending on phase

❖ Acute phase

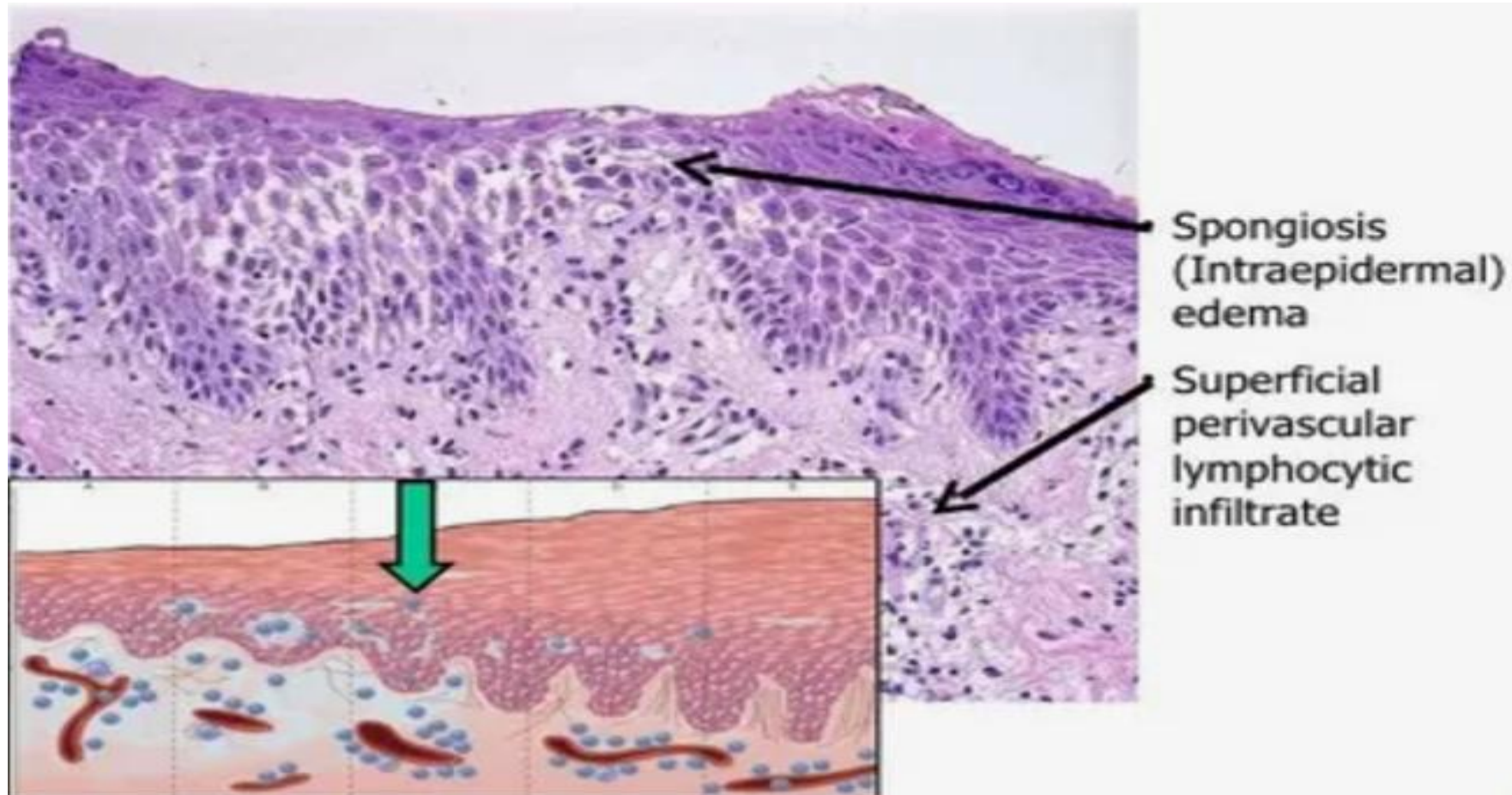
- shows **spongiosis** with **vesicle formation** in the epidermis

❖ Subacute phase

- shows less spongiosis and less acanthosis and hyperkeratosis

❖ Chronic phase

- shows acanthosis, hyperkeratosis and parakeratosis



Erythema Multiforme

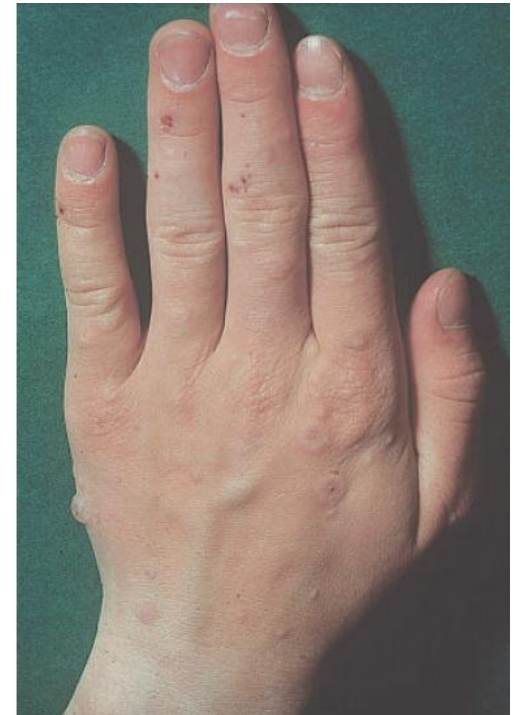
- ❖ An **acute, self-limited, immune-mediated** hypersensitivity reaction
- ❖ Characterized by its distinctive "multiform" skin lesions.
- ❖ It is primarily triggered by certain infections or drugs, leading to widespread **keratinocyte apoptosis**.

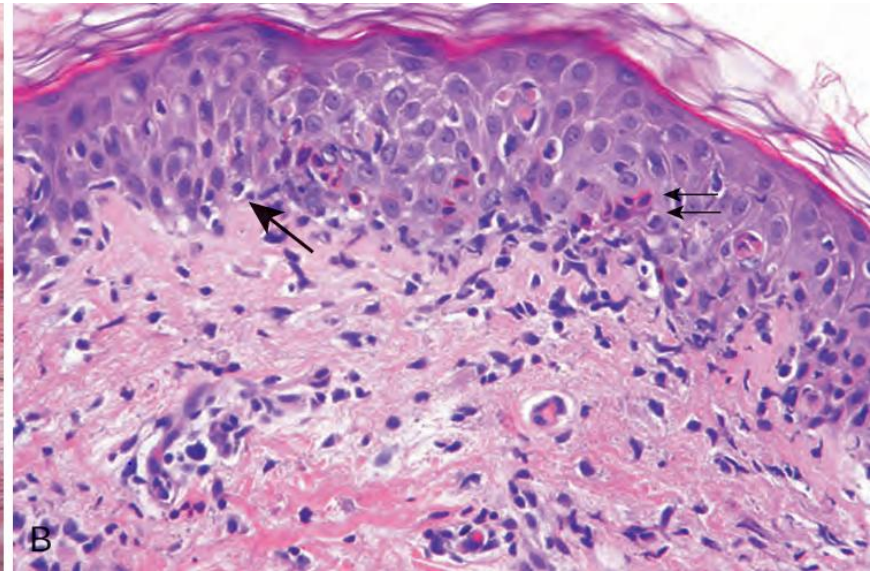
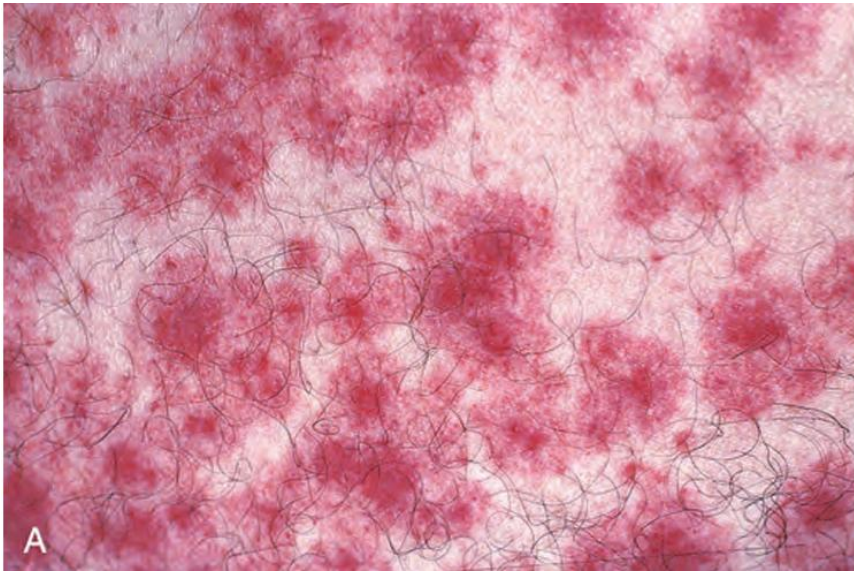
Etiology:

- ❖ **Infections (~90% of cases):** The most common trigger is **Herpes Simplex Virus (HSV)**. *Mycoplasma pneumoniae* is also a major cause, especially in children.
- ❖ **Drugs:** Less common (e.g., sulfonamides, anticonvulsants, NSAIDs).
- ❖ **Idiopathic:** A significant number of cases have no identifiable trigger.

Clinical feature

- ❖ Multiform lesion with multiform shapes including macules, plaques, vesicles, bullae
- ❖ Characteristic **Target lesion**- a pink border, and pale vesicular, eroded center.
- ❖ Often **acral** (distal extremities), with a symmetric distribution, frequently involving the **palms and soles**.
- ❖ If it extensively involves the mucosal surfaces with associated fever it is called **Steven-Johnson syndrome**
- ❖ **Toxic epidermal necrolysis**- which shows extensive skin and mucosal sloughing
- ❖ **SJS/TEN** is a separate, severe, and often **drug-induced** disease.





Chronic Inflammatory Dermatoses

- These last for months and years

Includes:

- Psoriasis
- Lichen planus
- Lupus erythematosus
- Acne vulgaris

Psoriasis

- ❖ A common chronic inflammatory disorder
- ❖ Clinically involves the **extensor surfaces** of the elbow & knee, lumbosacral area, inter-gluteal cleft, scalp
- ❖ Lesions have **silver white scale** which has underlying pink plaque
- ❖ Rubbing uninvolved skin induce a new lesion (**Koebner phenomenon**)
- ❖ Scratching the skin produces multiple small hemorrhages (**Auspitz sign**)
- ❖ **Nail changes** are seen in **30%** of cases and include yellowish discoloration, pitting, detachment of the nail from its base (onycholysis)
- ❖ **5%** develop psoriatic **arthritis**



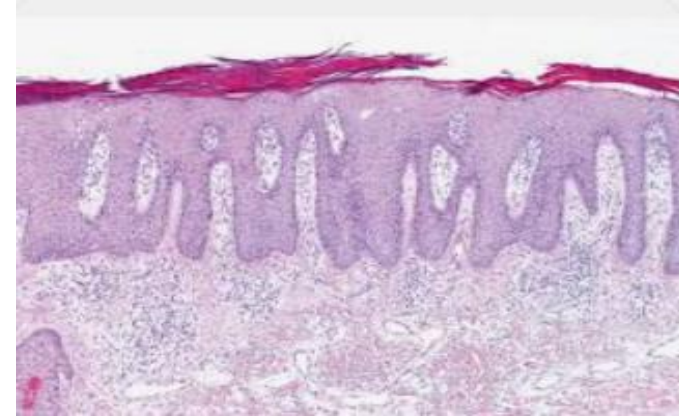
Pustular psoriasis

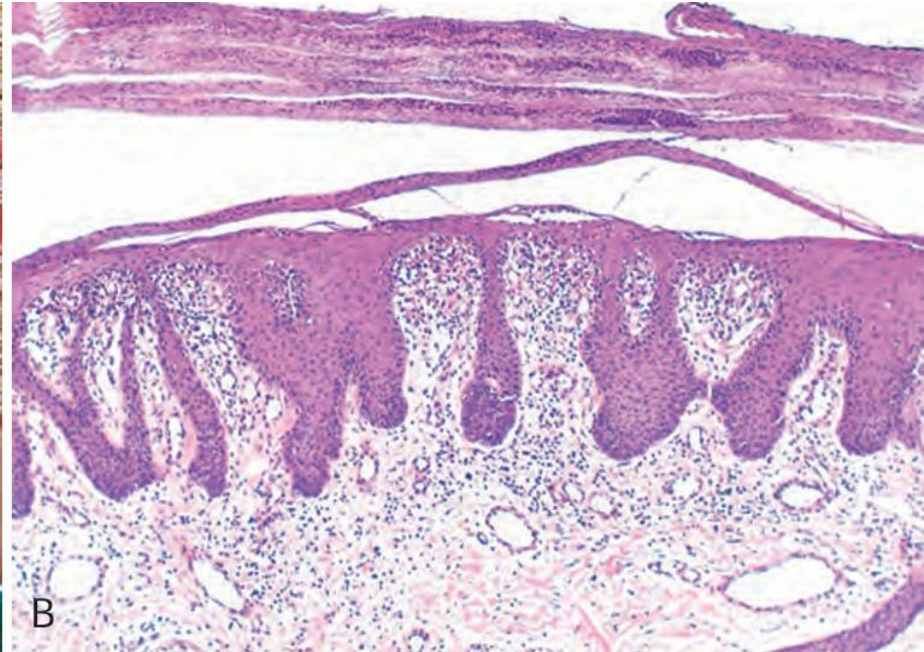
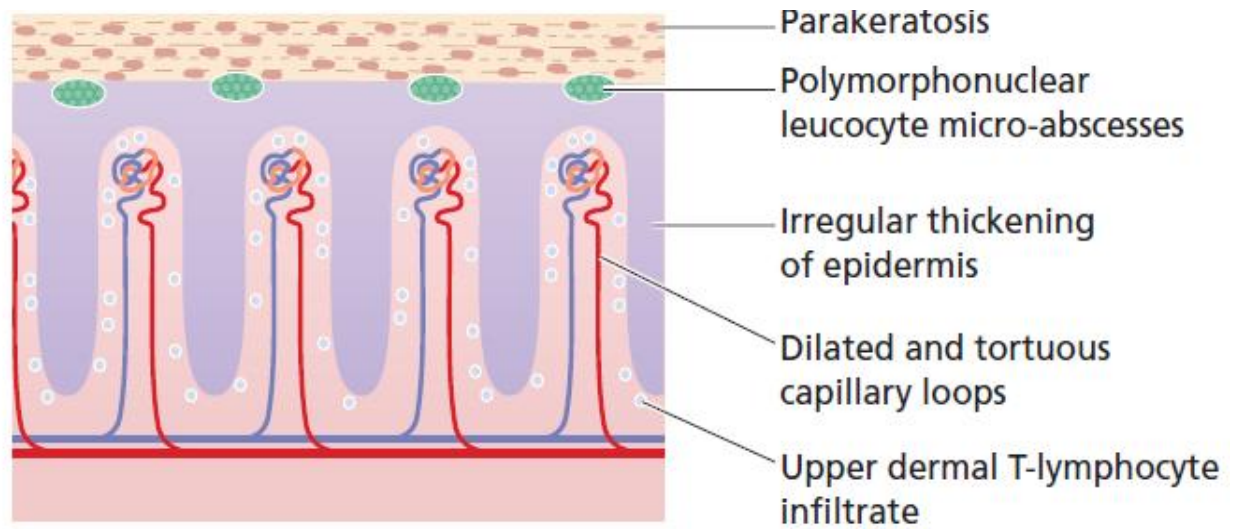
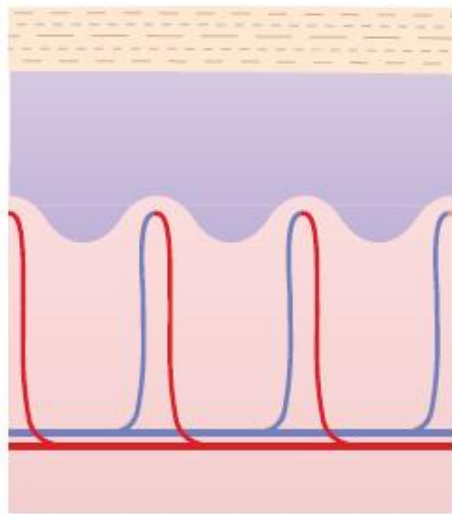
- ❖ Multiple pustules in the lesion
- ❖ This can be localized or generalized type which might be life threatening
- ❖ Psoriasis can cause erythroderma (generalized erythema and scaling of the lesion)



Microscopy

- ❖ Acanthosis
- ❖ Elongation of rete ridges
- ❖ Parakeratosis
- ❖ Munro's abscess (neutrophil aggregates in stratum corneum)
- ❖ Absent stratum granulosum
- ❖ The epidermis overlying dermal papillae is thinned (scratching it produces Asupitz sign)





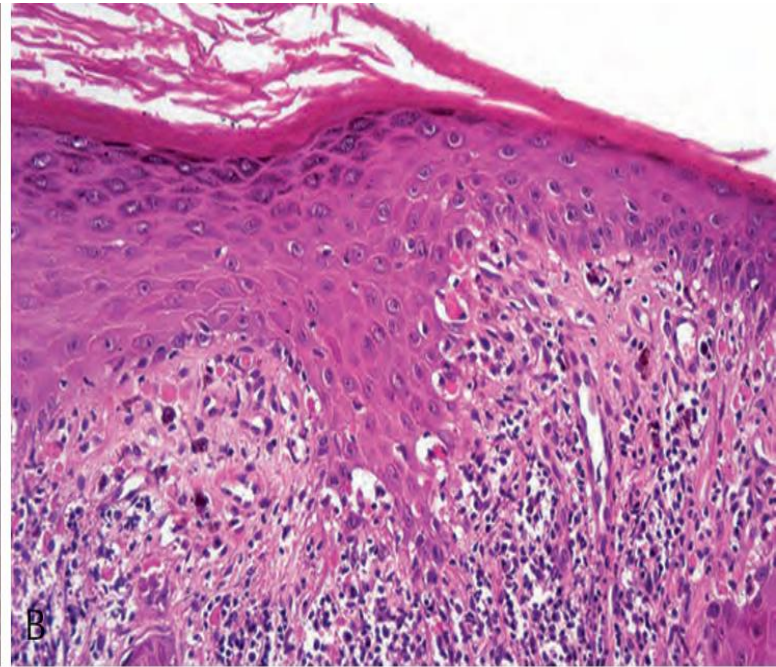
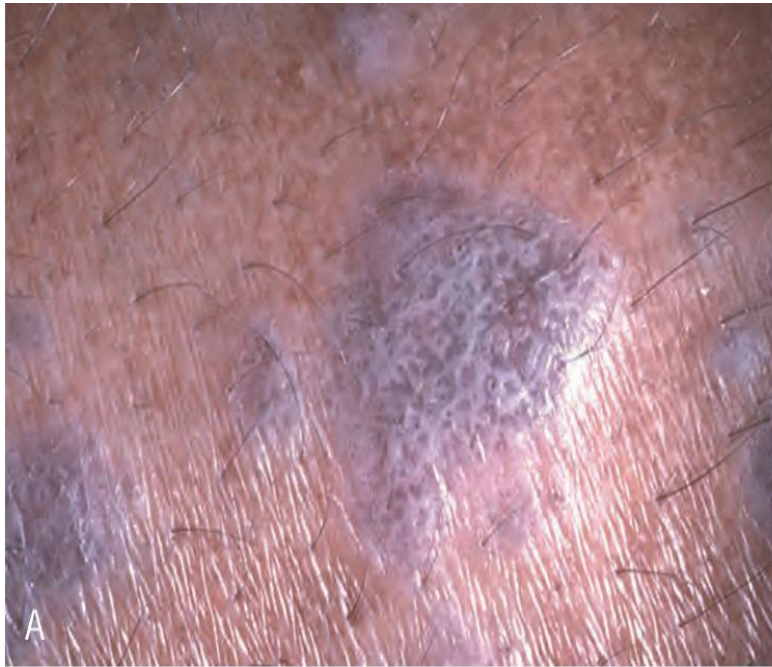
Pathogenesis of Psoriasis

- Genetic factors play a role
- Ag-Ab complexes in stratum corneum may lead to activation of the complement system, chemotaxis of neutrophils, release of growth factors that stimulate proliferation of epidermal cells.
- Lymphocytes are believed to be involved

Lichen Planus

- ❖ Present with 6 P's- Pruritic, purple, polygonal, planar, papules, and plaques”
- ❖ Involves the **wrists, bilaterally** and **symmetrically**
- ❖ It can also involve the **glans penis**
- ❖ Spontaneous resolution in **1 or 2 years**
- ❖ Microscopically shows basal cell destruction with lymphocytic infiltrates at dermoepidermal junction
- ❖ Cell mediated immunity of autoimmune nature





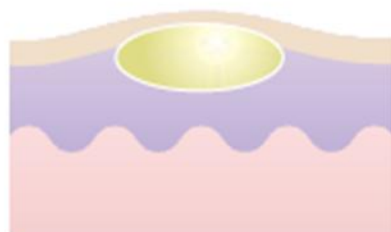
Bullous (Blistering) Skin Diseases

- ❖ Disorder with primary characteristic manifestation of **blister**.
- ❖ Classified into **two** main categories based on the histologic level of the split:
 - **Intraepidermal Blisters:** The cleavage plane is **within** the epidermis.
 - **Subepidermal Blisters:** The cleavage plane is **at or below** the **dermal-epidermal junction**
- ❖ The clinical appearance of the blister (e.g., **flaccid vs. tense**) often correlates with this level.

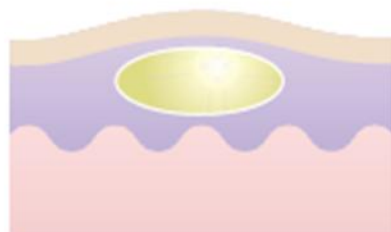
These include:

- Pemphigus vulgaris
 - Bullous pemphigoid
 - Dermatitis herpetiformis
 - Porphyrias
 - Epidermolysis bullosa
-
- ❖ Blisters can also be produced by other diseases such as acute eczematous dermatitis, HSV and Varicella zoster infection, etc.

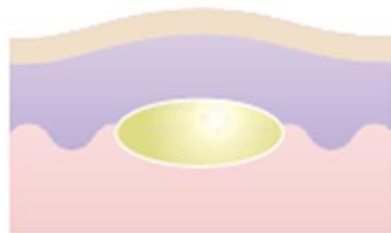
Location of bullae



Subcorneal bulla



Intra-epidermal bulla



Sub-epidermal bulla

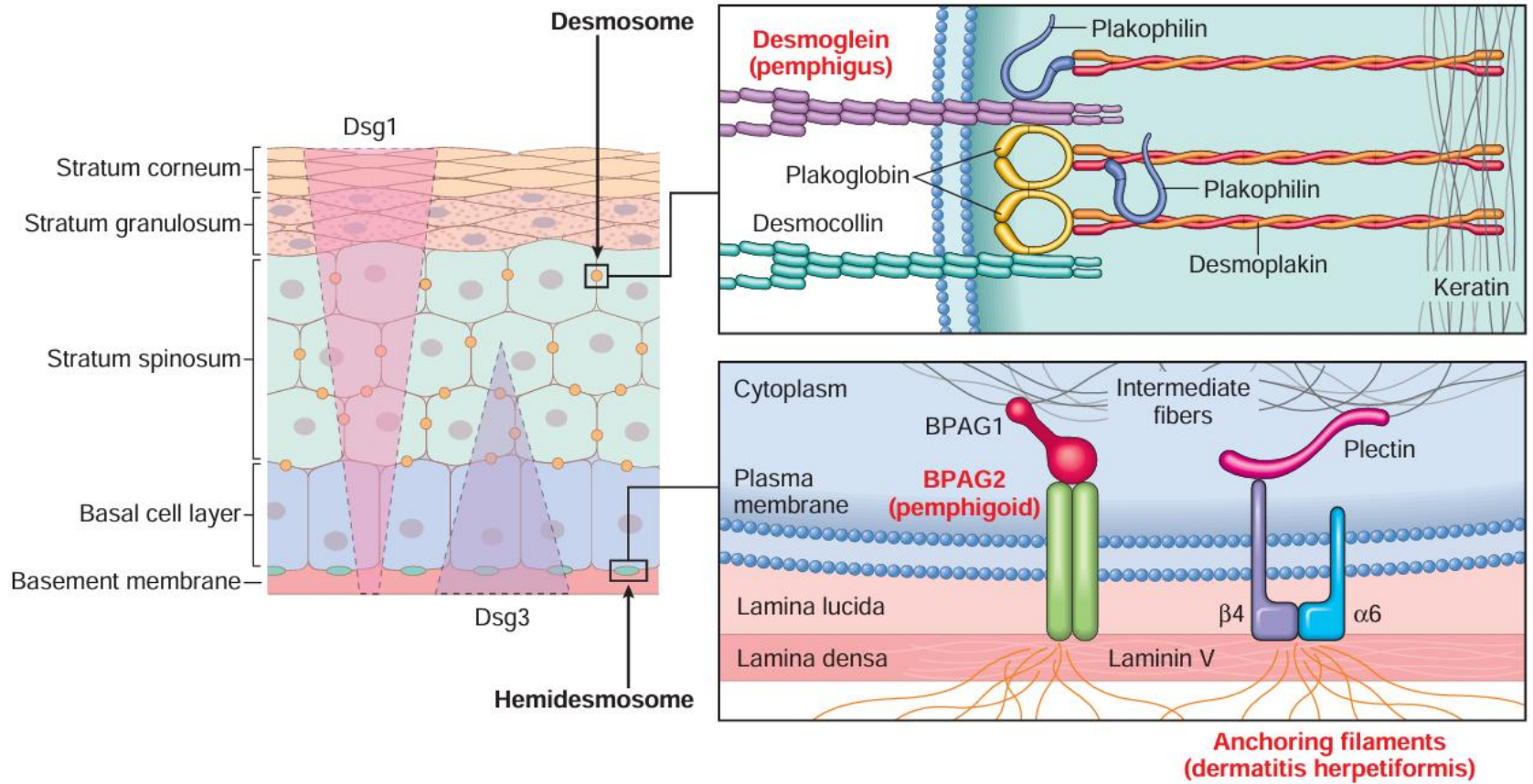
Diseases

Bullous impetigo
Miliaria crystallina
Staphylococcal
scalded skin syndrome

Acute eczema
Viral vesicles
Pemphigus
Miliaria rubra
Incontinentia pigmenti

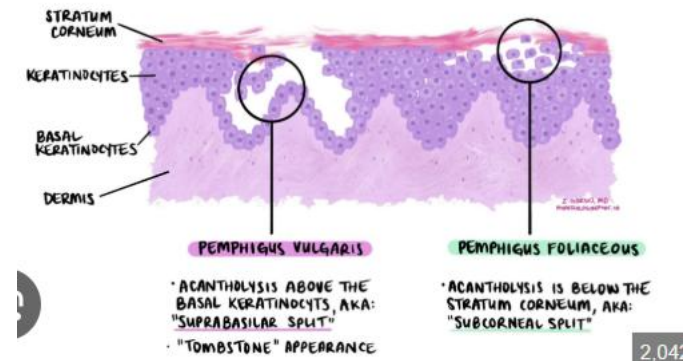
Bullous pemphigoid
Cicatricial pemphigoid
Pemphigoid gestationis
Dermatitis herpetiformis
Linear IgA disease
Bullous erythema multiforme
Bullous lichen planus
Bullous lupus erythematosus
Porphyria cutanea tarda
Toxic epidermal necrolysis
Cold or thermal injury
Epidermolysis bullosa





Pemphigus

- An autoimmune disease with **IgG Abs** directed against mucous membranes and skin
- This leads to acantholysis, and blister formation
- 4th-6th decade
- 4 subtypes:
 - **Pemphigus vulgaris**
 - Pemphigus vegetans
 - **Pemphigus foliaceus**
 - Pemphigus erythematosus



Pemphigus vulgaris

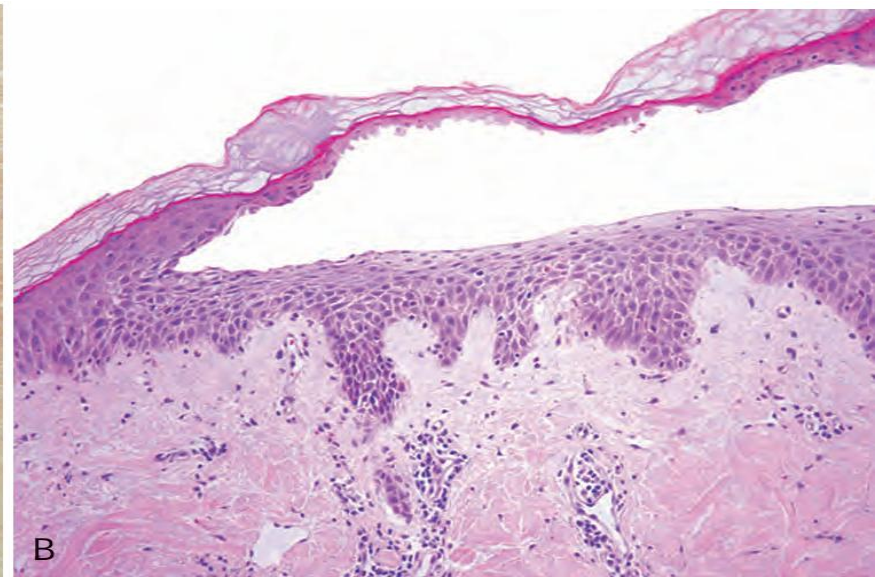
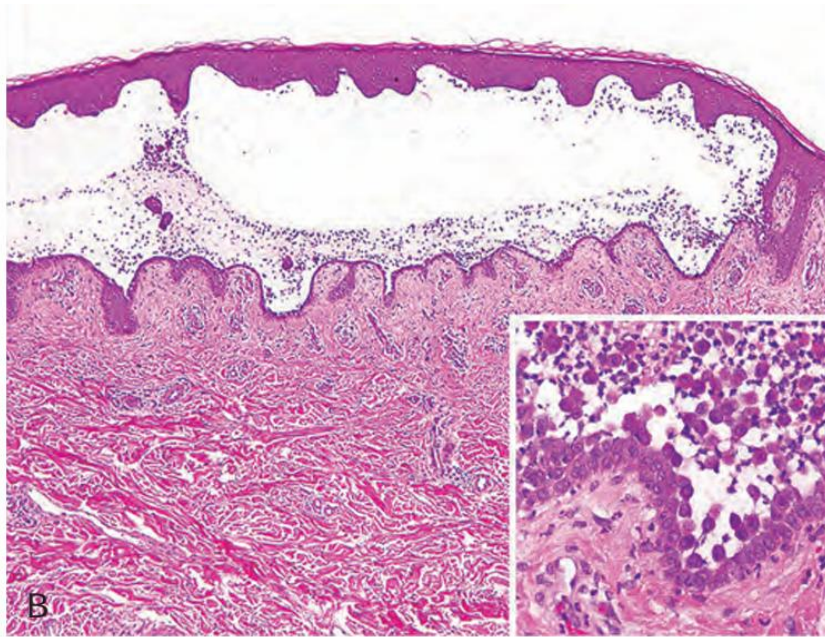
- ❖ Shows vesicles in the scalp, face, axilla, trunk, groin and mucous membranes
- ❖ The roof of the vesicle is **thin** and ruptures easily and become crusted

Microscopy

- acantholysis and interepidermal vesicle formation.
 - It is not subepidermal.

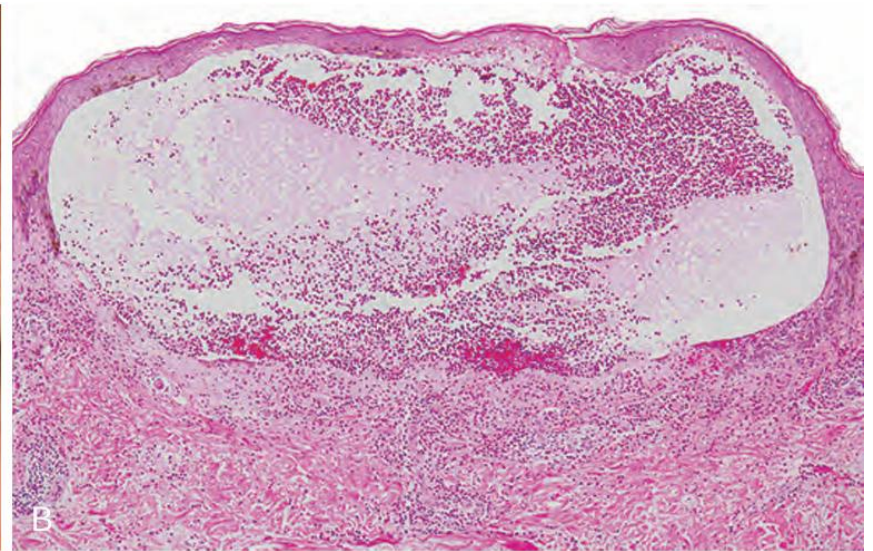
❑ **Pemphigus vegetans** has **suprabasal** vesicles and **foliaceus** has vesicles in the **upper** most epidermis.

❖ **DIM:- IgG (fishnet pattern)**



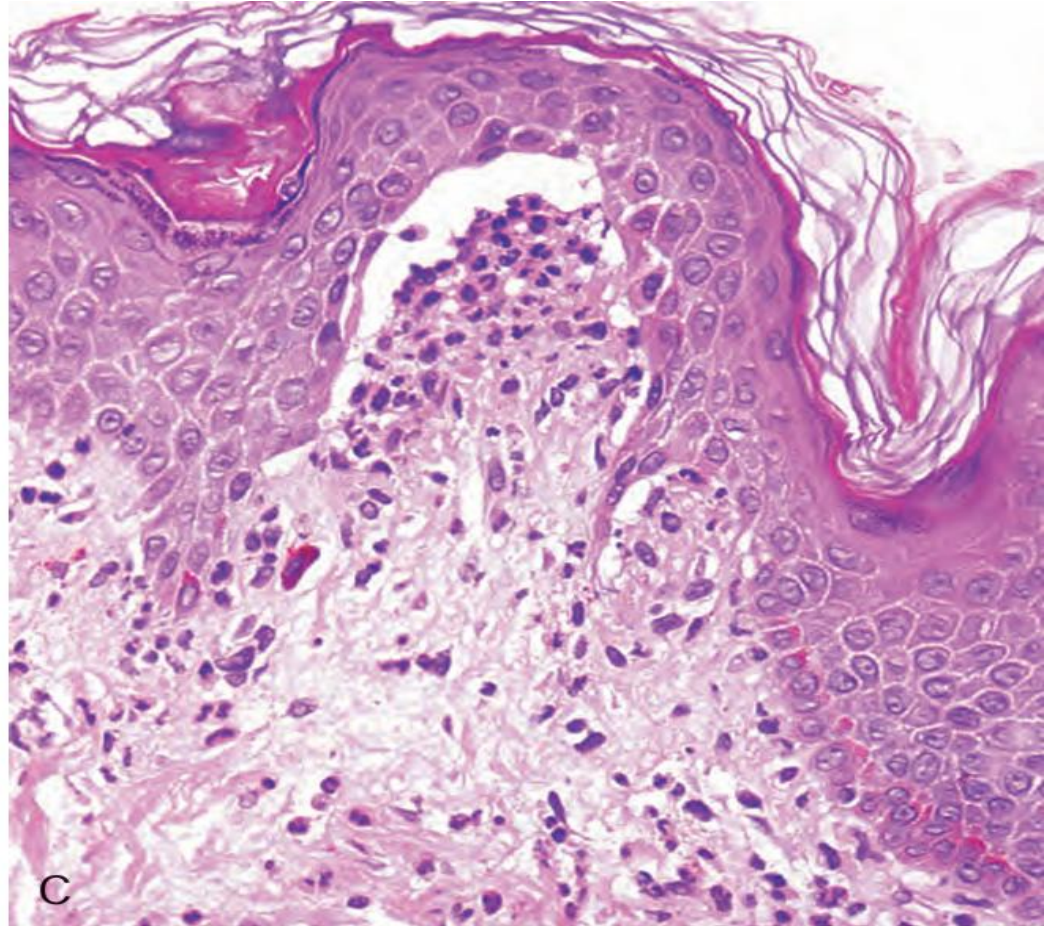
Bullous Pemphigoid

- ❖ An **autoimmune disease** with auto antibodies directed against **basement membrane** of epidermis
 - IgG target **BPAG2**
- ❖ Commonly occurs in the **elderly**
- ❖ Shows intact **tense vesicles** which do not rupture b/c the blister is formed **subepidermally**
- ❖ Primarily on **flexural** areas
- ❖ DIF:- Shows a **linear deposition of IgG and C3** along the basement membrane zone.



Dermatitis Herpetiformis

- ❖ Rare disorder
- ❖ Clinically manifests with recurrent **pruritic blisters**
- ❖ The blisters are **clustered** like herpes simplex infection
- ❖ Usually located on skin of **knees, elbows, upper back, buttock**
- ❖ Microscopy shows **neutrophilic microabscesses** at the **tips of the dermal papillae**, which eventually coalesce to form blisters.
- ❖ Can be associated with **Celiac disease**
 - **IgA** antibodies against dietary gluten cross-react with **epidermal transglutaminase** in the skin.
- ❖ **DIF**: Shows **granular deposits of IgA** in the dermal papillae.



Epidermolysis Bullosa

- ❖ Hereditary disorders showing blisters immediately after birth
- ❖ Caused by mutations in genes encoding structural proteins that anchor the epidermis to the dermis.
 - It is **not autoimmune**.
- ❖ Usually at sites of pressure, trauma or rubbing.

Acne Vulgaris

- ❖ Due to altered maturation of the hair follicles.
- ❖ It can be non inflammatory or inflammatory or combined
- ❖ Accumulation of keratin and sebum in the hair follicles
- ❖ This may protrude through the opening of hair follicles producing **comedon** which has a **white color**
- ❖ Accumulated keratin can rupture and induce acute & chronic inflammation in the surrounding
- ❖ Resolution of inflammation is followed by scar formation
- ❖ Acne has uncertain pathogenesis and genetic & hormonal factors may play a role



Panniculitis

- Inflammation of the **subcutaneous fat**.

Subtypes:

- Erythema nodosum (>80% of cases)
- Erythema induratum
- Weber-Christian disease
- Factitious panniculitis

Erythema nodosum:

- ❖ Clinically presents with tender nodules usually in the lower extremities
- ❖ presents acutely

Etiology:

- β -hemolytic strep, Tb, leprosy, histoplasmosis, coccidioidomycosis
- Drugs- OCP
- Malignancies
- Sarcoidosis, inflammatory bowel diseases
- Idiopathic



Infections and Infestations

- Skin diseases can be caused by microorganisms and insects

Molluscum contagiosum

- Caused by **pox virus**
- Transmitted by direct contact
- Can be seen in children & adults
- Self limiting lesion causing **papules with umblicated center**
- Microscopically a cup shaped epidermal hyperplasia with molluscum cytoplasmic inclusion bodies.



Verrucae (warts)

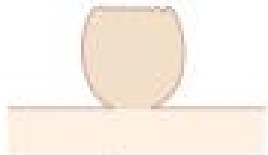
- Caused by HPV
- Usu. spontaneously regress in 6 months to 2 years
- Based on clinical morphology and location:
 - Verruca vulgaris usu occurs in the dorsum of the hand and fingers as a papule
 - Verruca plantaris- occurs in sole of feet
 - Verruca palmaris- occurs in the palm
 - Verruca plana (flat wart)
 - Condyloma accuminatum as a cauliflower like lesion over the penis and female genitals.
- Microscopically all verrucae have a fibrovascular core lined by thickened epithelium showing koilocytosis
- Different HPV subtypes cause different warts



Plane (flat) warts are common on hands and faces of children



A mosaic wart is made of many single warts packed together in a plaque. Always hard to treat



Cauliflower and digitate patterns are most common in the beard and genital areas



Warts may appear in lines after the skin has been cut or scratched (the Köbner phenomenon)



Other infections and infestations

- HSV
- Herpes zoster
- HPV
- CMV
- Tinea (Dermatophytosis)-
 - superficial skin infections by dermatophytes.
 - Infection is localized to the skin.
 - Various names given depending on the site of body involved.
- Arthropods

Impetigo

- ❖ Is a common superficial bacterial infection of skin.
- ❖ It is highly contagious and is frequently seen in otherwise healthy children.
- ❖ Usually involves exposed skin, particularly that of the face and hands.
- ❖ Two forms exist,
- ❖ Classically referred to as impetigo contagiosa and impetigo bullosa
 - *S. aureus*
- ❖ Presents as an erythematous macule, but multiple small pustules rapidly supervene.
 - honey-colored crust
- ❖ Accumulation of neutrophils beneath the stratum corneum



Premalignant and Malignant Epidermal Tumors

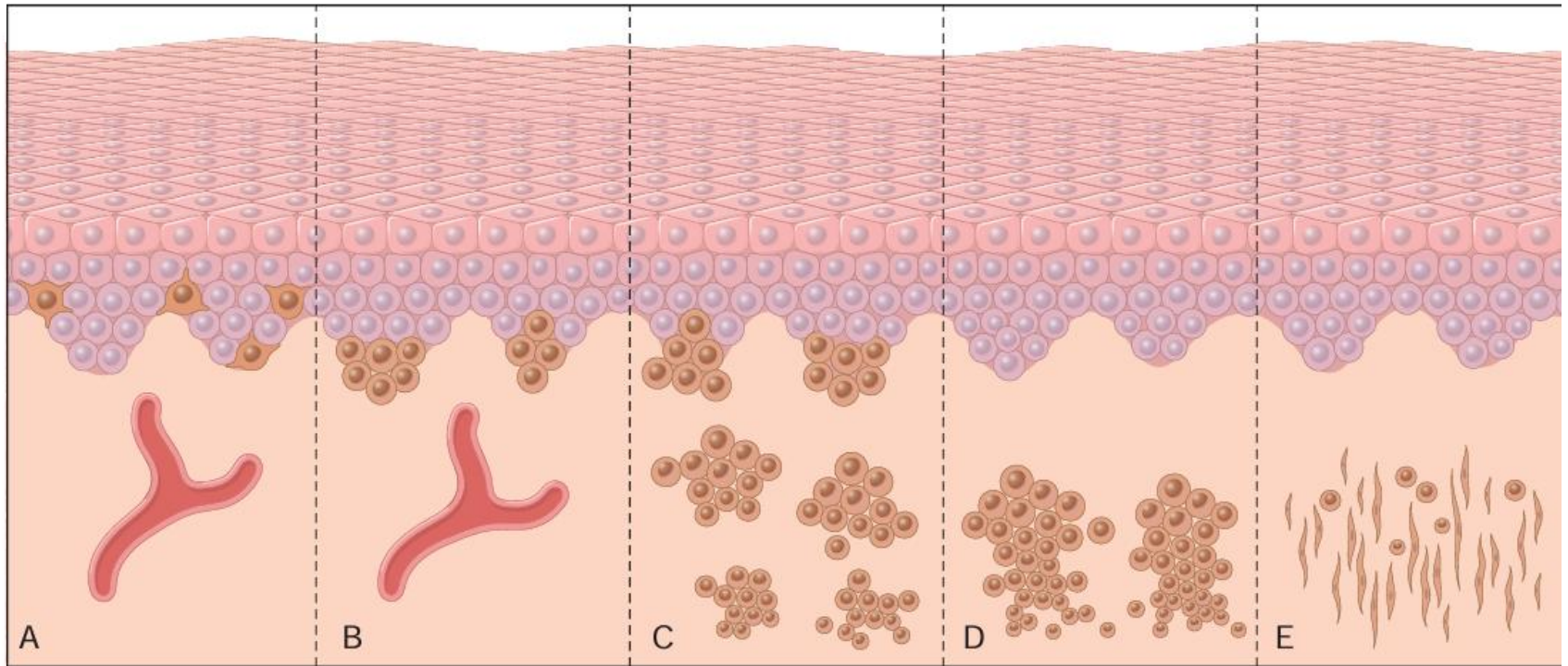
- ❖ These are among the most common human cancers
- ❖ Primarily induced by **cumulative UV radiation exposure.**

Melanocytic Nevi (nevocellular nevi, moles)

- ❖ Benign melanocytic tumors
- ❖ Brown or black uniformly pigmented papules or nodules that have a smooth round border
- ❖ Different clinical and histological subtypes:
 - Common acquired nevus
 - Congenital nevus
 - Spitz nevus
 - Halo nevus
 - Blue nevus
 - Dysplastic nevus
- ❖ Some nevi can progress to malignant melanoma
 - Large **congenital nevi** and **dysplastic nevi** have increased risk

Table 25.2 Representative Variant Forms of Melanocytic Nevi

Nevus Variant	Diagnostic Architectural Features	Cytologic Features	Clinical Significance
Congenital nevus	Deep dermal and sometimes subcutaneous growth around adnexa, neurovascular bundles, and blood vessel walls	Identical to ordinary acquired nevi	Present at birth; large variants have increased melanoma risk
Blue nevus	Non-nested dermal infiltration, often with associated fibrosis	Highly dendritic, heavily pigmented nevus cells	Black-blue nodule; often confused with melanoma clinically
Spindle and epithelioid cell nevus (Spitz nevus)	Fascicular growth	Large, plump cells with pink-blue cytoplasm; fusiform cells	Common in children; red-pink nodule; often confused with hemangioma clinically
Halo nevus	Lymphocytic infiltration surrounding nevus cells	Identical to ordinary acquired nevi	Host immune response against nevus cells and surrounding normal melanocytes
Dysplastic nevus	Coalescent intraepidermal nests	Cytologic atypia	Potential marker or precursor of melanoma



Malignant Melanoma

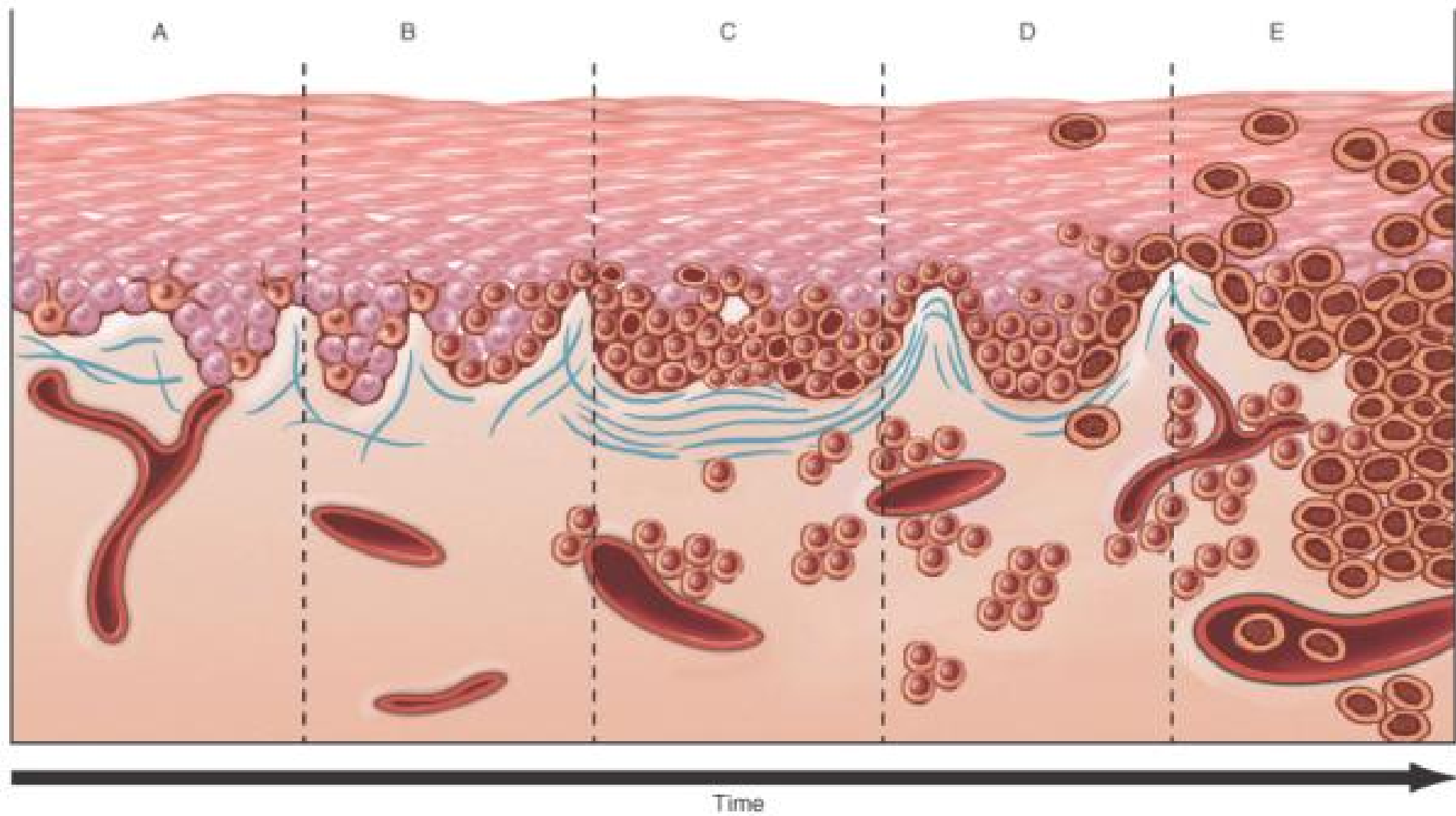
- ❖ The most deadly of all skin cancers
- ❖ Strongly linked to acquired mutations caused by exposure to UV radiation in sunlight
 - **CDKN2A**
 - Increases in RAS and PI3K/AKT signaling
 - TERT
- ❖ Occurs commonly in older people with fair skin
- ❖ Predisposing factors
 - Excessive sunlight
 - Prior melanocytic or dysplastic nevi
 - Genetic predisposition
- ❖ Phases of growth:
 - Radial growth phase
 - Vertical (nodular) growth phase

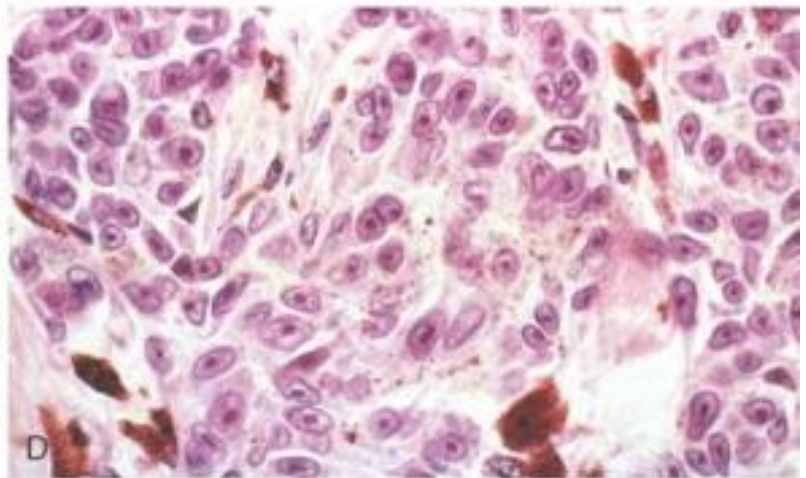
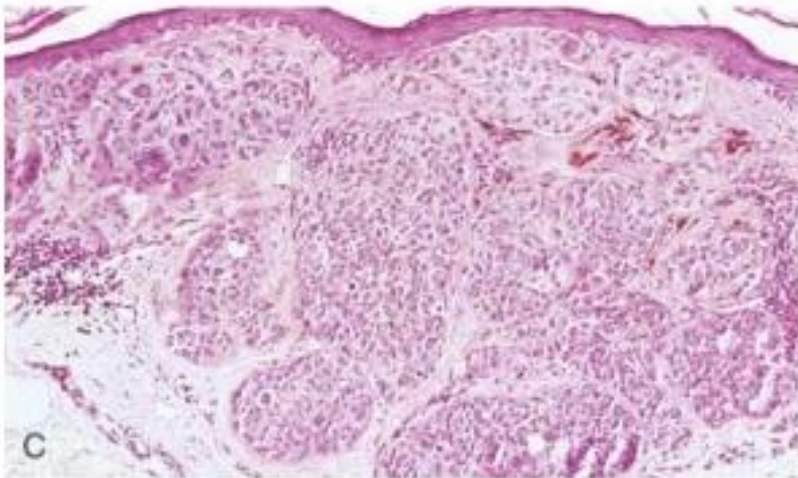
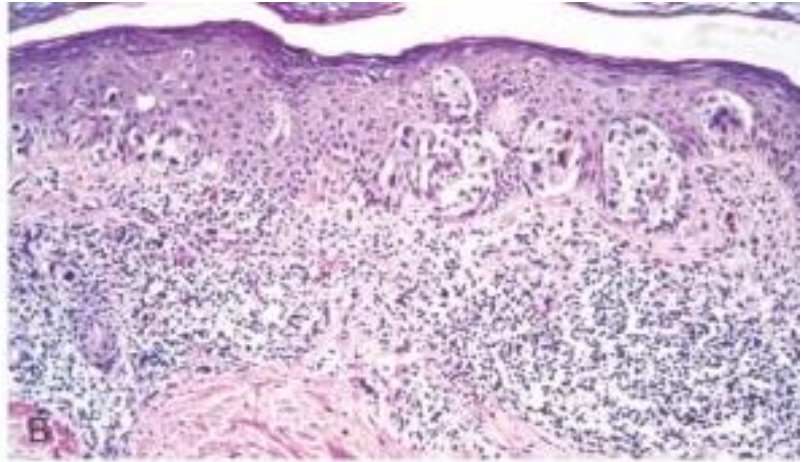
Radial/initial/horizontal growth phase

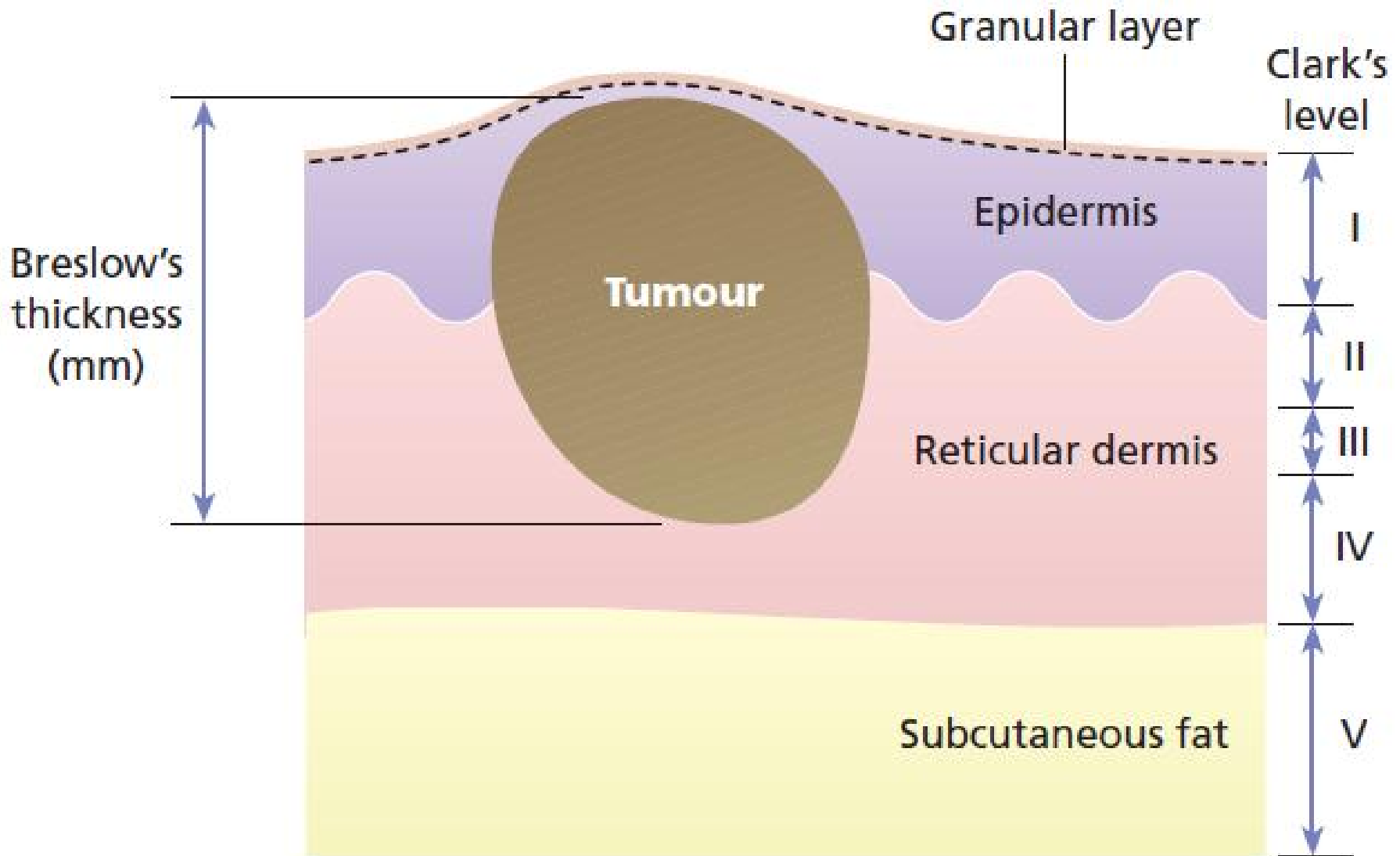
- ❖ Proliferation of malignant melanocytes in the **epidermis and upper part of dermis.**
- ❖ Does **not** have the capacity to invade
 - Can not metastasize at this stage
- ❖ Clinically appears as **hyperpigmented macules**
- ❖ Subtypes:
 - **Lentigo maligna melanoma**
 - occurs in the face of the elderly
 - When it's converted to vertical growth phase, the change is heralded by the appearance of a **nodule**
 - **Superficial spreading melanoma**
 - occurs in the trunk and the extremities.
- ❖ These lesions can be cured by simple excision.

Vertical growth phase

- ❖ Show clusters of malignant melanocytes in the **dermis**.
- ❖ It has capacity for **invasion and metastasis**.
- ❖ Clinically indicated by appearance of a **nodule** in a previous macule
 - **Acral lentiginous** melanoma is an exception (it's progression is not heralded by a nodule)
- ❖ The risk of metastasis is directly proportional to the depth of invasion.
 - The deeper the invasion, the higher the risk of metastasis





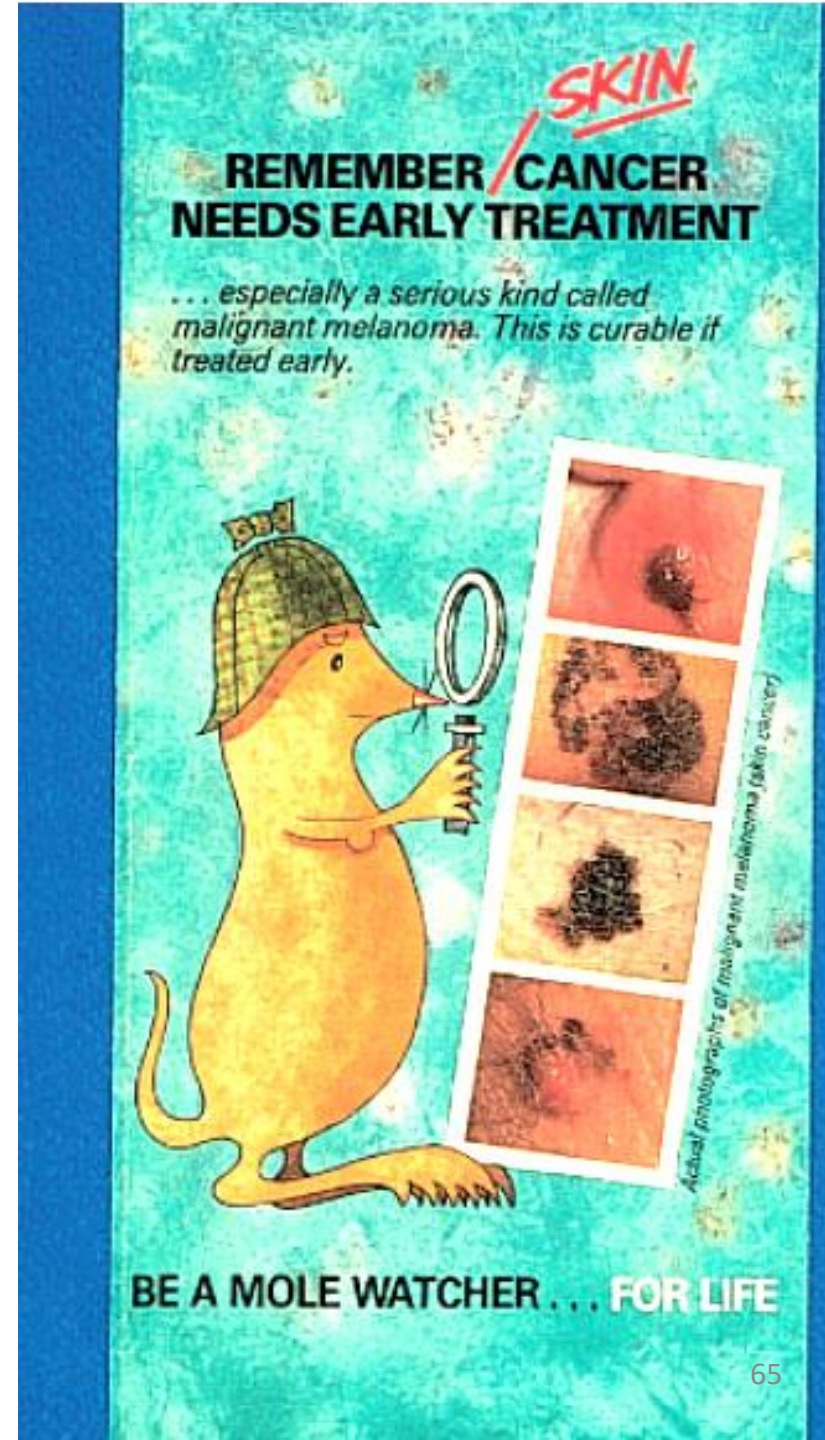


Clinical feature

- Hyperpigmented macule or nodule which shows variegation in color, irregular and angulated border, painful or itchy



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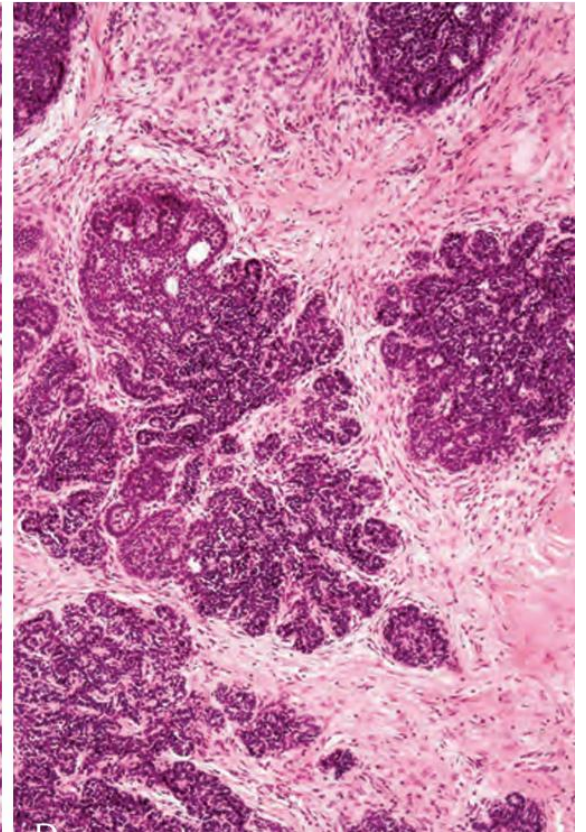
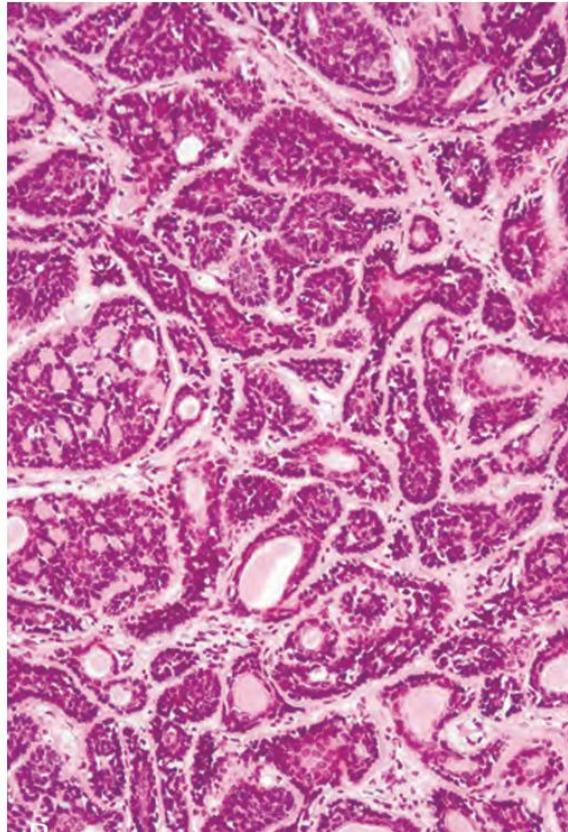


Benign Epithelial Tumors

- There are many forms and should be distinguished from malignant tumors
 - Seborrheic keratosis
 - Acanthosis nigricans
 - Fibroepithelial polyp (skin tag, acrochordon)
 - Epithelial cysts
 - Epidermal inclusion cyst
 - Dermoid cyst
 - Pilar cyst (Trichilemmal cyst)
 - Sebaceous adenoma – arises from sebaceous glands

Tumors of skin adnexae

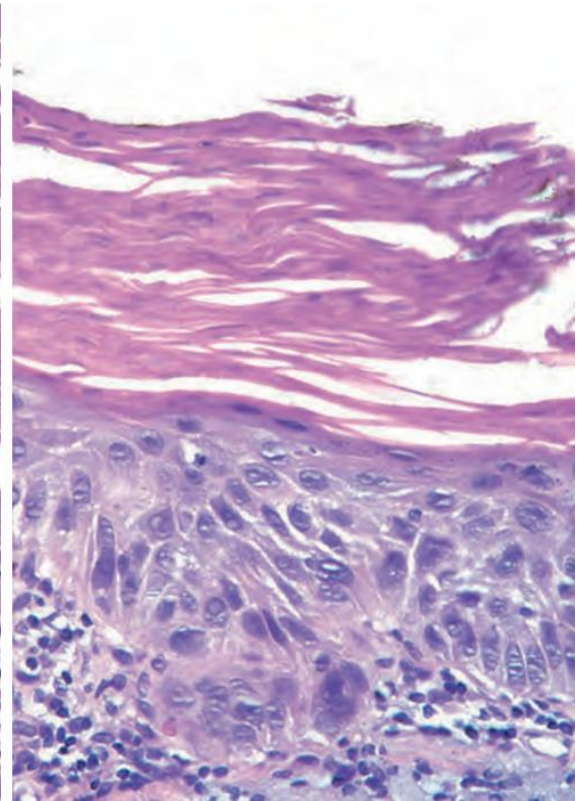
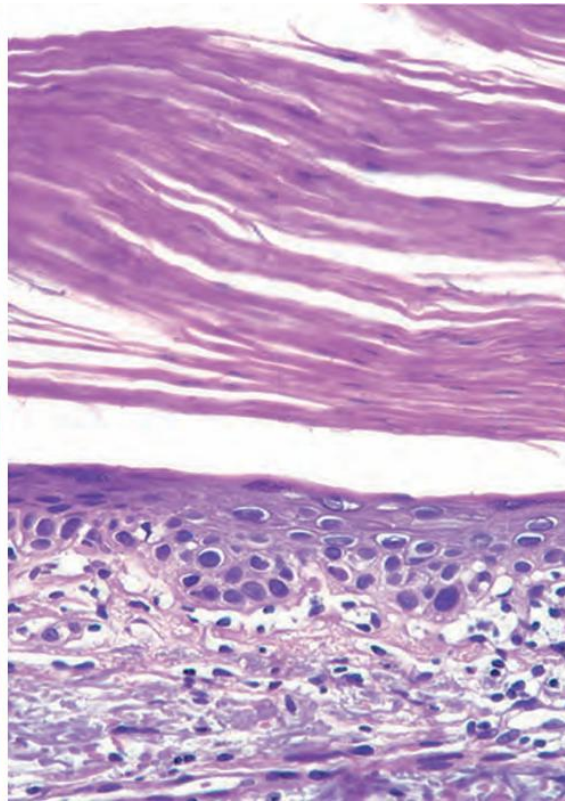
- Benign tumors of skin adnexae
 - Cylindroma- arises from **apocrine/eccrine** glands
 - Hidradenoma- arises from **apocrine** glands
 - Syringoma- arises from **eccrine sweat** glands
 - Eccrine poroma
 - Trichilemmoma
 - Trichoepithelioma
 - Trichofolliculoma
- Skin adnexae can give rise to malignant tumors
 - Apocrine ca, eccrine ca, sebaceous ca



Premalignant and Malignant Epidermal Tumors

Actinic Keratosis

- ❖ A premalignant lesion commonly occurring in **elderly** individuals
- ❖ Caused by **excessive sunlight exposure**
- ❖ Sun exposed skin- face, arms, dorsum of the hand
- ❖ Appears as **scaly plaque** with poorly demarcated borders.
- ❖ Occasionally **excess keratin** production produces a **horn-like structure**
- ❖ Microscopy shows **basal cell atypia**



Squamous cell carcinoma

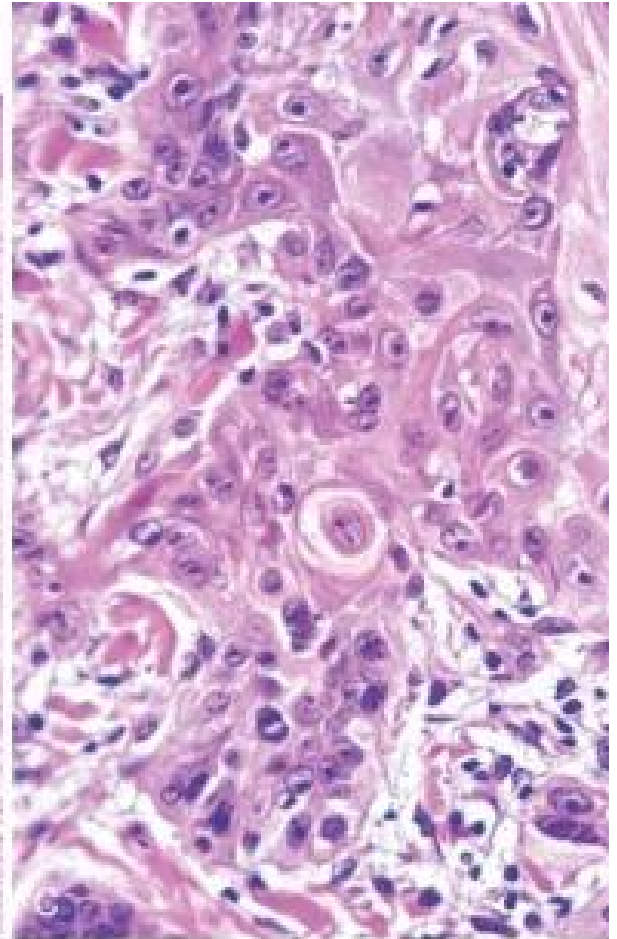
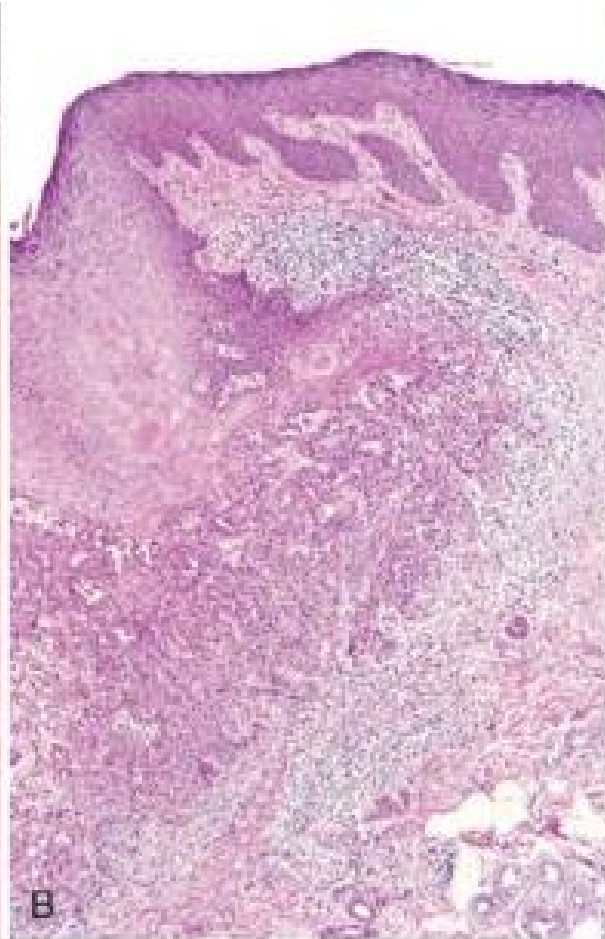
- ❖ A common skin tumor
- ❖ DNA damage induced by exposure to UV light
 - **TP53 mutations**
- ❖ Predisposing factors:
 - **Excess UV exposure**
 - Ionizing radiation
 - **Xeroderma pigmentosum**
 - Industrial toxins (tar and oils)
 - Chronic ulcers
 - Chronic burn ulcers
 - Draining osteomyelitis
 - **Immunosuppression (chemotherapy)**
 - Tobacco and Bethel nut chewing (oral SCC)

Clinical features of SCC

- ❖ It can arise in the skin or mucous membranes
- ❖ In the skin most common sites include sun exposed regions (face, arms)
- ❖ SCC can follow actinic keratosis
- ❖ Appears as a **scaly plaque** or **ulcerated nodule**
- ❖ In the oral mucosa it appears as **leukoplakia** with whitish thickening

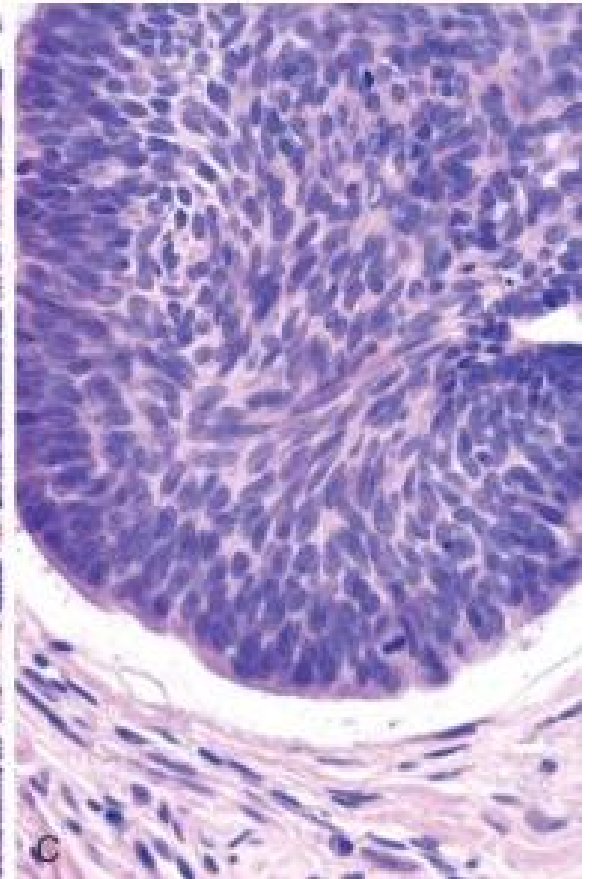
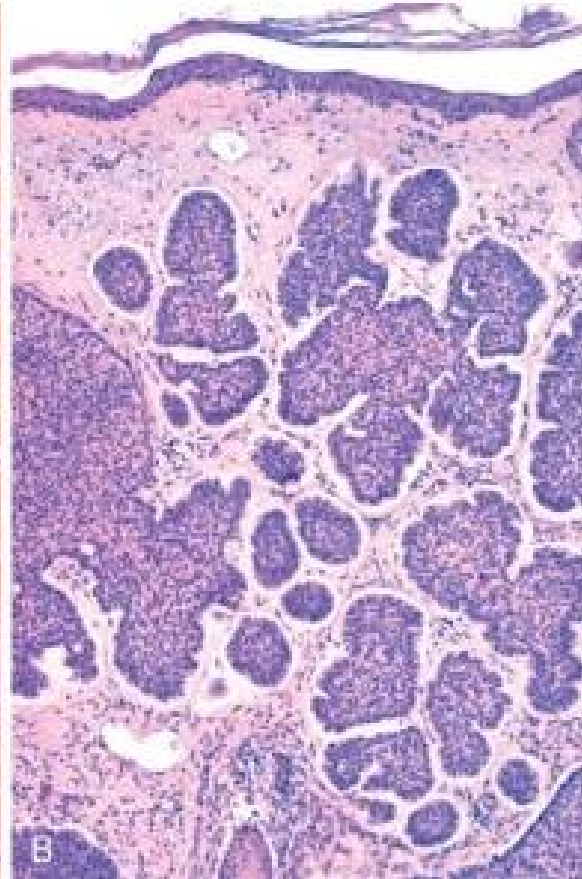
Microscopy

- ❖ **Carcinoma in situ - full thickness atypia in the epidermis**
- ❖ **Disruption of basement membrane - invasive SCC**
 - Nests and sheets of malignant squamous cells in the dermis with production of keratin
- ❖ SCC is a **locally invasive tumor** (<5% metastasize to regional lymph nodes)
- ❖ Rx- Excision of tumor



Basal cell carcinoma

- ❖ The most common invasive cancer in humans
- ❖ Associated with mutations that activate the **Hedgehog signaling pathway**
- ❖ More frequently occurs in light skinned individuals
- ❖ Predisposing factors include
 - excessive UV exposure
 - Immunosuppression
 - xeroderma pigmentosum
- ❖ Clinically starts as a papule over the usually in the head and neck
- ❖ If BCC is neglected it leads to **rodent ulcer**
- ❖ A locally aggressive tumor that **rarely metastasizes**
- ❖ Microscopically shows **basaloid cells** in clusters with **peripheral palisading**.
- ❖ Rx- excision of tumor



Dermal Tumors

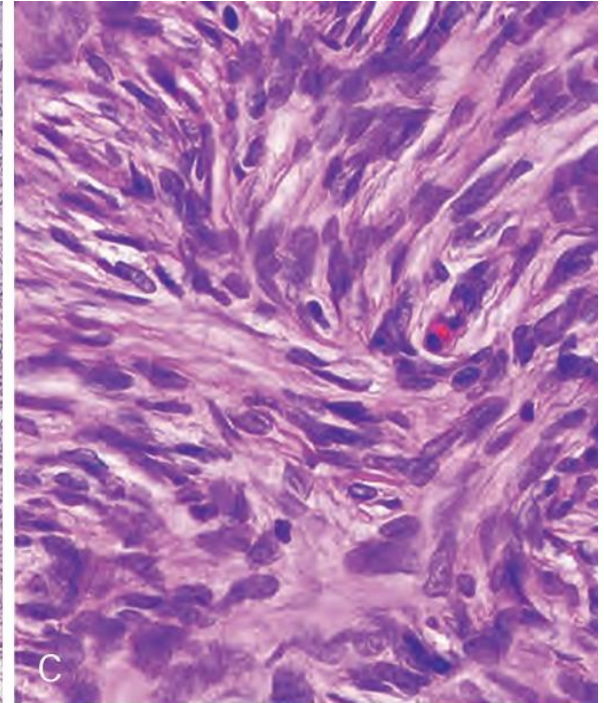
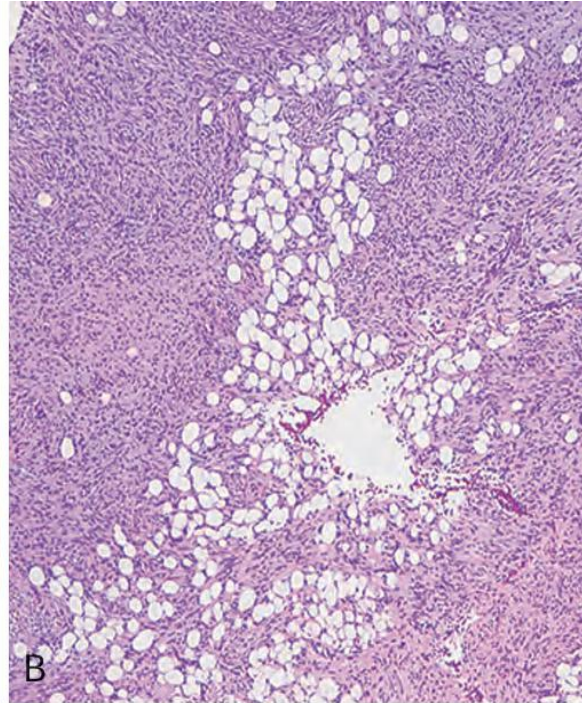
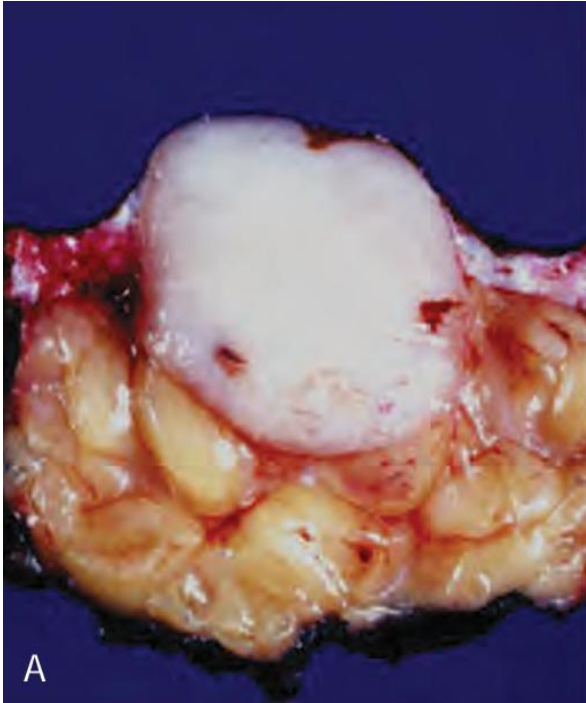
- ❖ Dermis contains fibroblasts, blood vessels, lymphatics, nerves, smooth muscle
- ❖ Tumors can arise from these structures (benign and malignant)

Benign Fibrous Histiocytoma (Dermatofibroma)

- benign dermal neoplasm of uncertain lineage.
- usually seen in adults and often occur on the legs
- appear as firm, tan to brown papules

Dermatofibrosarcoma Protuberans

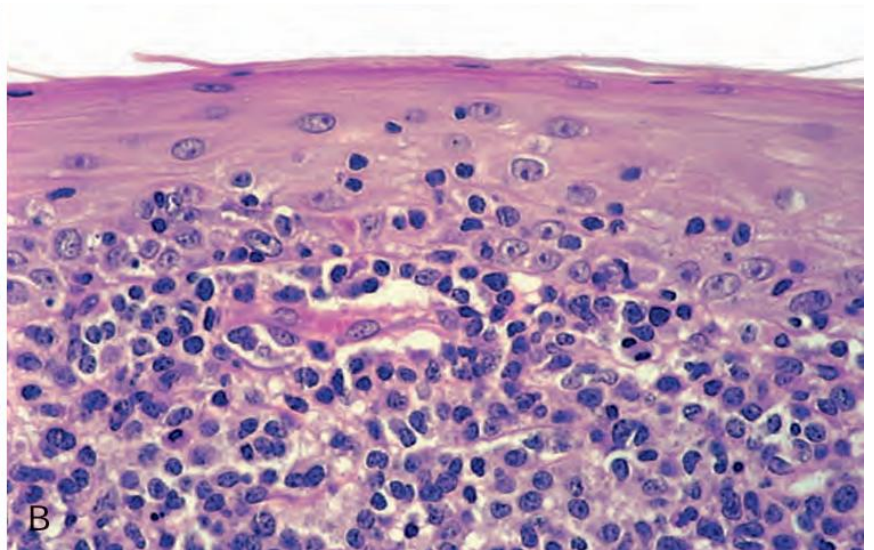
- well-differentiated, primary fibrosarcoma of the skin
- locally aggressive and can recur, they rarely metastasize



Tumors of cellular immigrants

- Langerhans cells - Histiocytosis X
- Mast cells- Mastocytosis
- Mycosis Fungoides
 - lymphoma of skin-homing **CD4+ T-helper cells**
 - most commonly afflicts persons **older than age 40**
 - involve **truncal** areas
 - Patch, plaque and nodule
 - diffuse erythema and scaling of the entire body surface (erythroderma): **Sézary syndrome**
 - Other types of lymphomas can arise in the skin

Cutaneous T-cell lymphoma



Disorders of Pigmentation

❖ Albinism

❖ Vitiligo .

❖ Melasma

❖ Lentigines

Albinism

- ❖ Hereditary disorder causing failure of production of melanin by melanocytes.
- ❖ May be due to deficiency of thyroxinase
- ❖ Two forms are recognized:
 - Oculocutaneous albinism
 - Cutaneous albinism



Vitiligo

- Acquired disorder leading to **loss of melanocytes** causing skin **hypopigmentation**
- Thought to be an **autoimmune** process
- May be seen along with other autoimmune diseases like Graves disease and pernicious anemia



Melasma

- ❖ Hyperpigmentation of **facial skin**
- ❖ Commonly occurs during **pregnancy** but disappears after delivery
- ❖ Due to **increased melanin transfer** to keratinocytes and dermal macrophages

Lentigo (lentigines)

- ❖ Hyperpigmented skin lesions usually seen in children
- ❖ Benign localized linear proliferation of melanocytes above the basement membrane